

Investigations of binding energy for SPP using the ANL-Osaka DCC model in NEUT



- ▶ Introduction to NEUT Single-Pion Productions
- ▶ Implementation of the new DCC model to NEUT
- ▶ Investigation of nuclear models

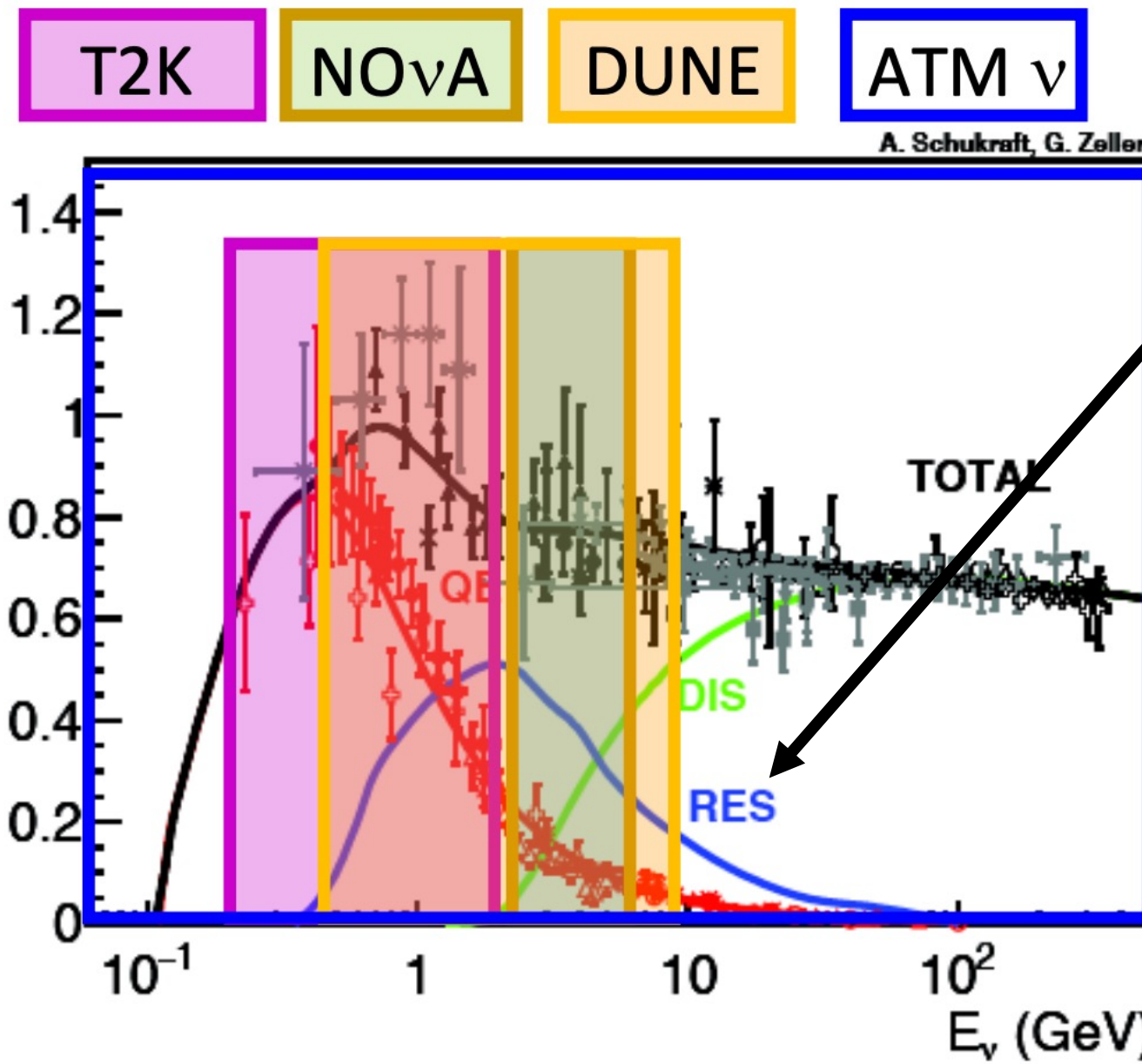


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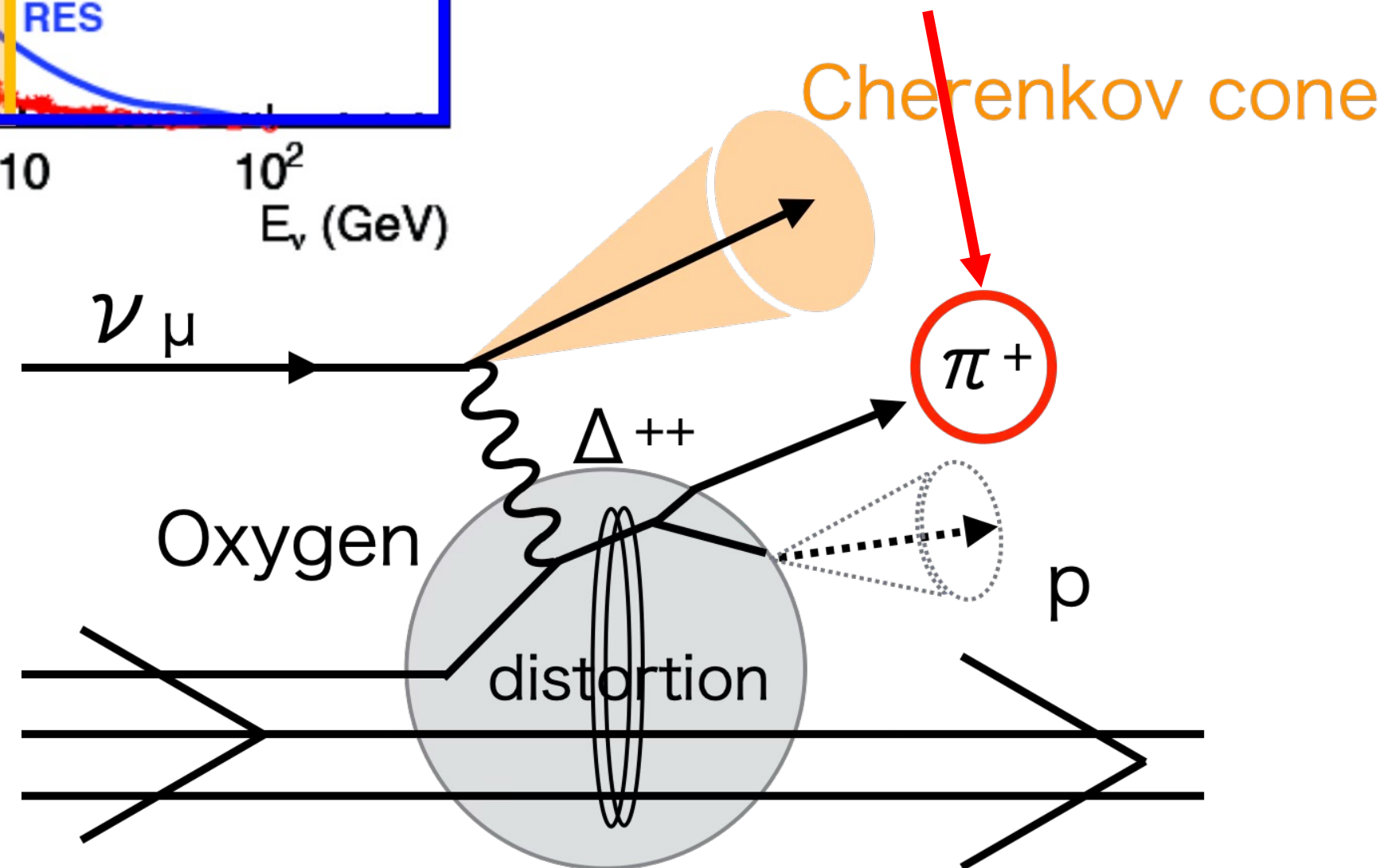
- This talk is focuses on DCC model in NEUT.
- Other activities for SPP (Single-pion prouction) in NEUT, such as HNV and MK implementation, would be given by Clarence tomorrow.

sub-dominant channel in ν osc. exp.

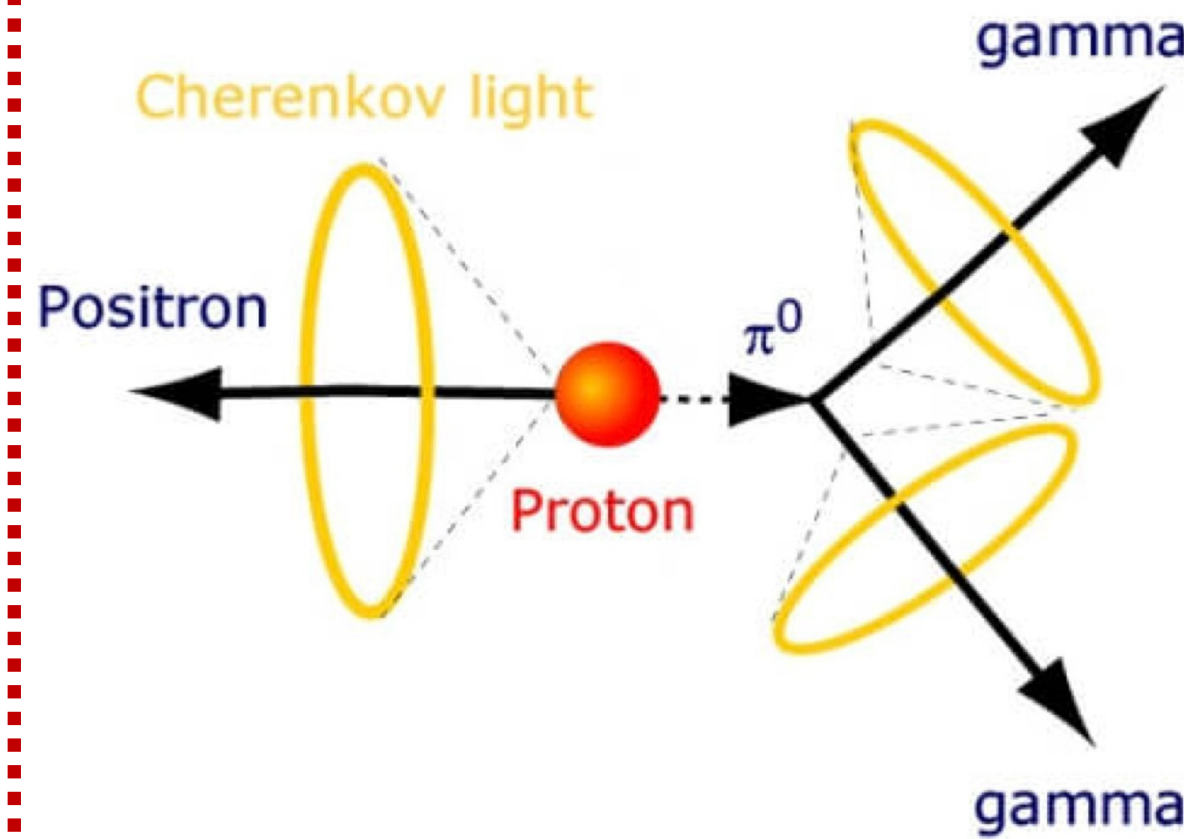


SPP

Key: π
absence/presence,
and its kinematics

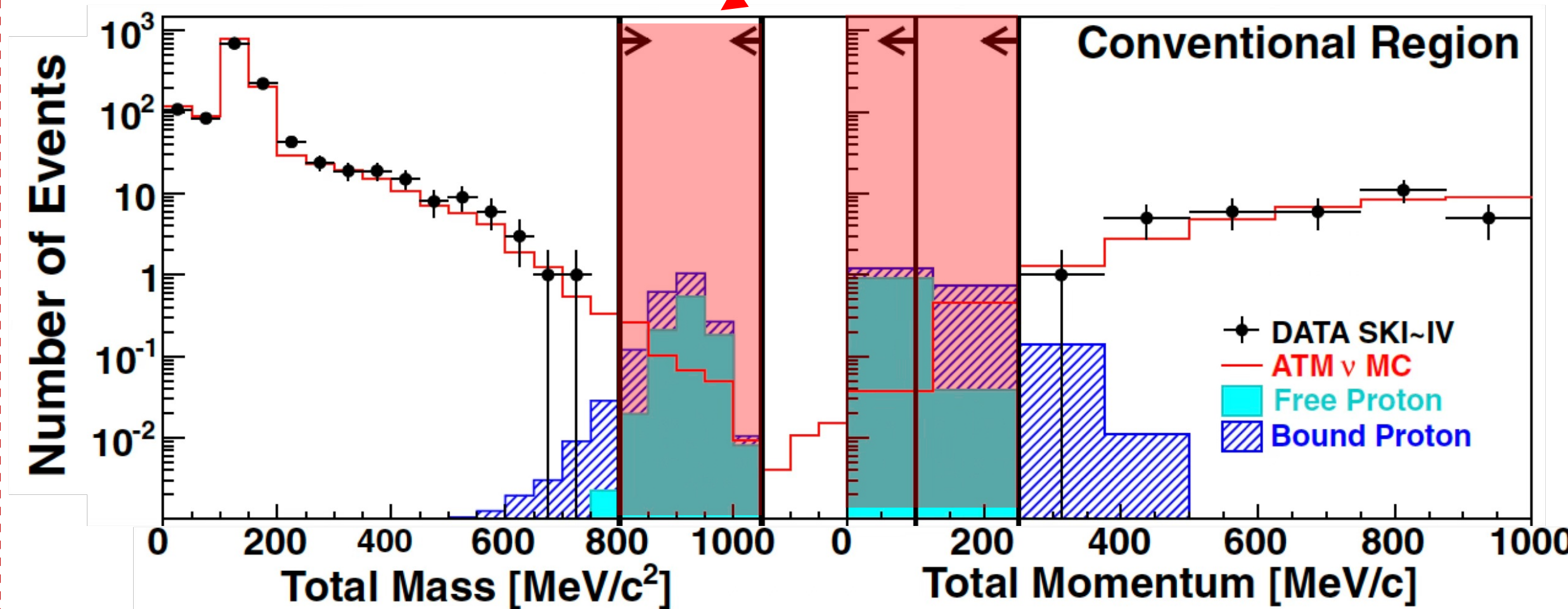


Major BG in nucleon decay search.



Select “back-to-back”
events, but atm. ν BG
remains.

signal region



Partially adapted from Phys.
Rev. D 102, 112011 (2020)

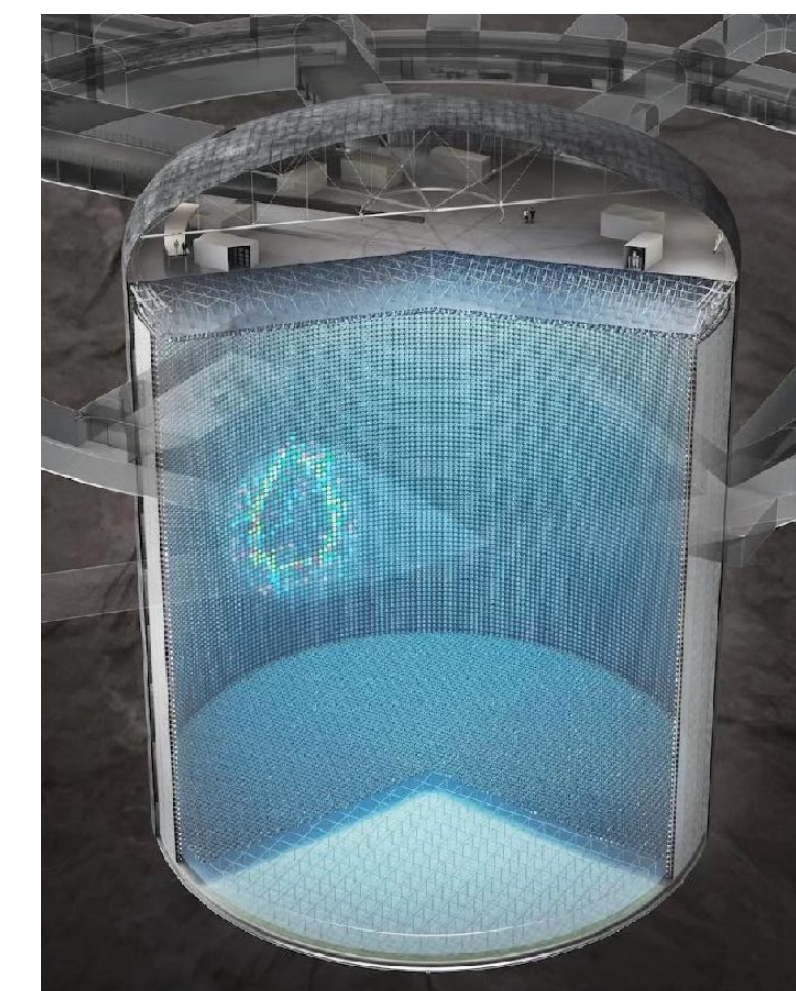
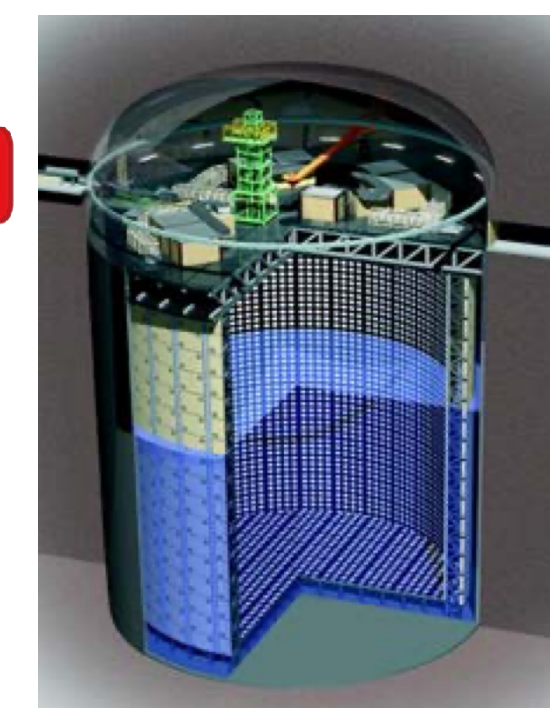
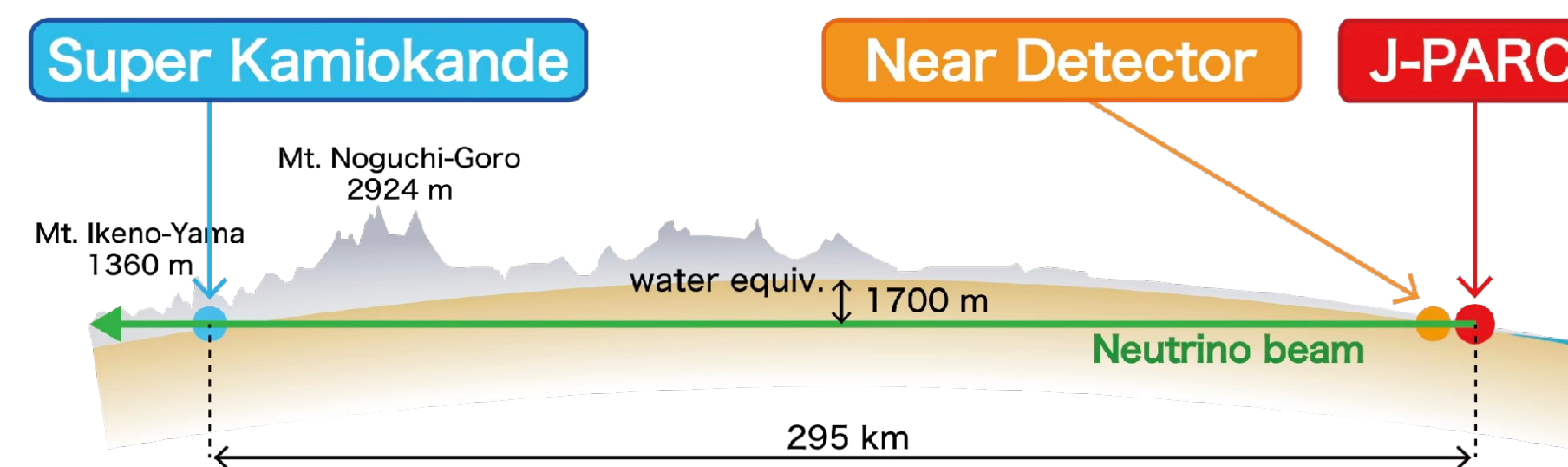
- Neutrino experiments are entering into a high-precision era.
 - c.f.) Hyper-Kamiokande, DUNE, JUNO, etc.
- Precise prediction of SPP by event generator is required.**

NEUT neutrino event generator

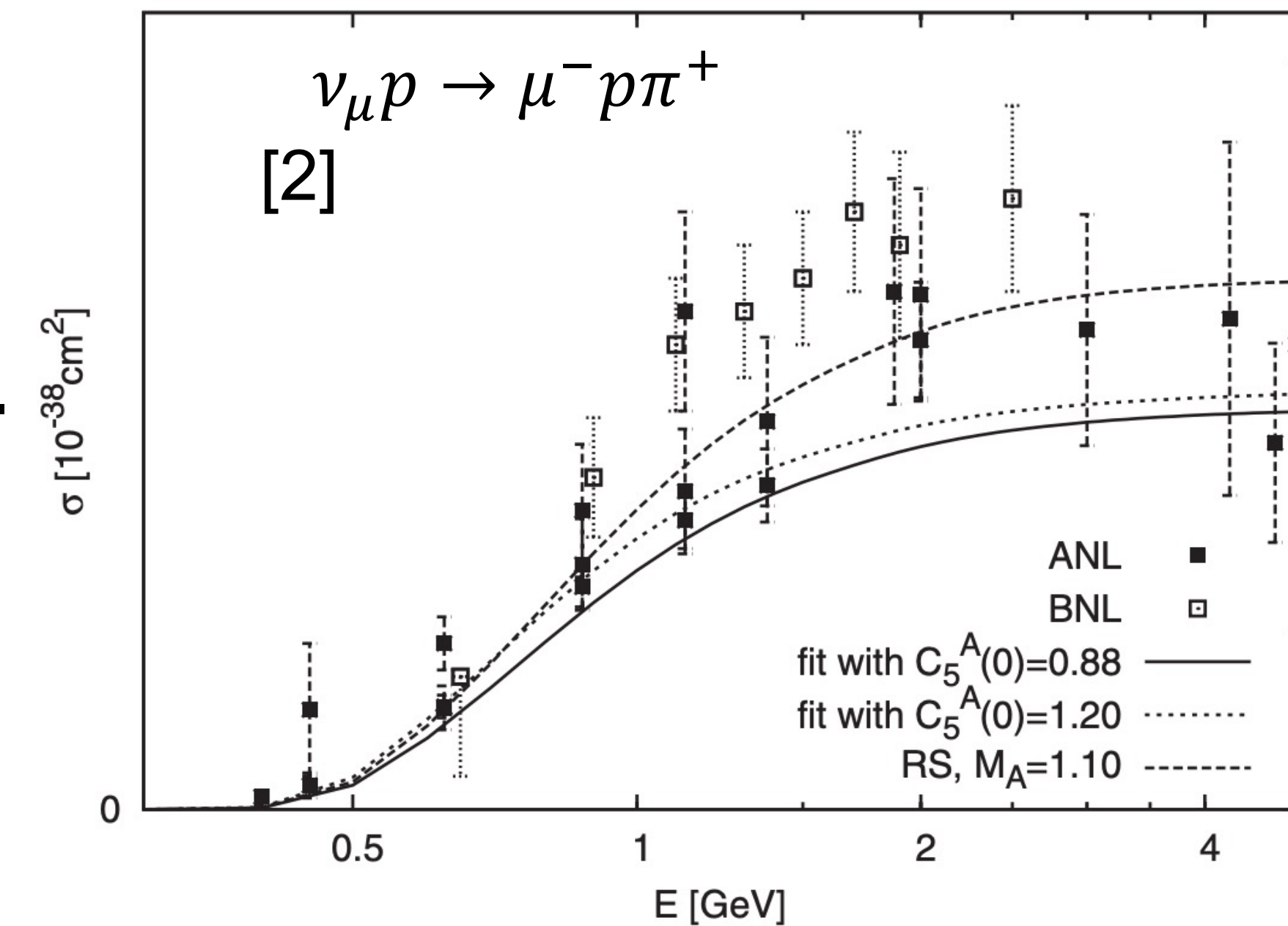
- ▶ Born for the Kamiokande experiment (NOT Super-K) in the 1980s.
- ▶ Mainly used in experiments in Japan: Super-K and T2K.
 - The primary target is **water**, but it can be used for carbon, iron, etc.
- ▶ Not open to public yet.
 - Contact us if you are interested in.
 - We have started discussion to make it public among developers.
- ▶ Aiming for Hyper-K and other next-generation neutrino experiments, we keep developing NEUT.

Y. Hayato and L. Pickering,
Eur. Phys. J. Special Topics 230, 4469 (2021).

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(Creation Date and Author)
1983.??.?? ; M.NAKAHATA
1987.08.?? ; N.SATO   FOR TAU
1988.08.31 ; T.KAJITA  DATA UPDATE
1988.09.06 ; T.KAJITA  R1314 IS ADDED
```



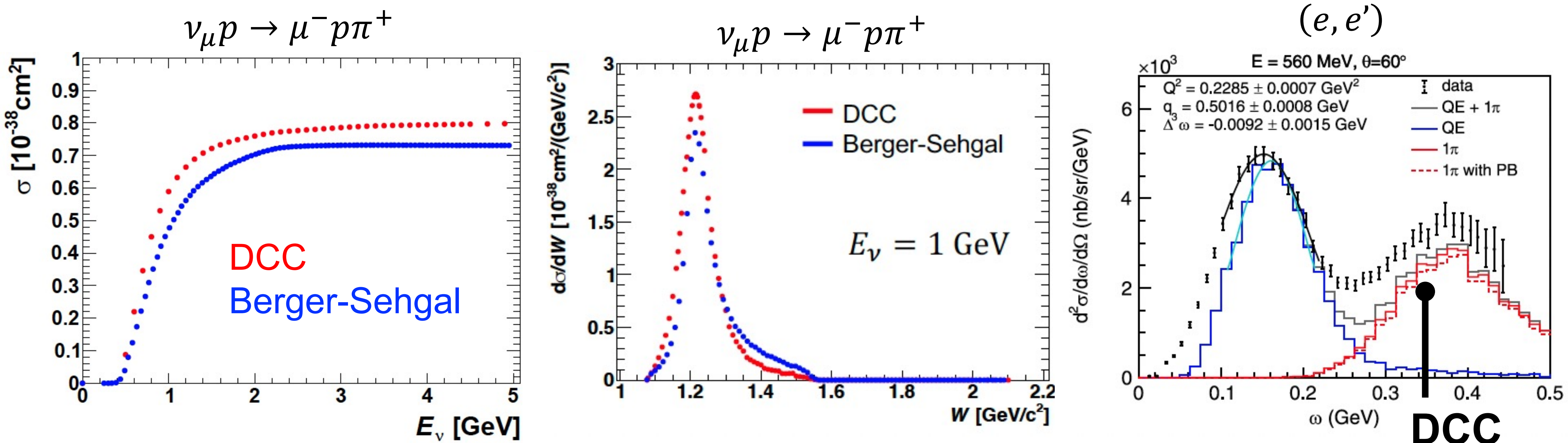
- ▶ Rein-Sehgal (RS) model has been used for long-time.
 - with Graczyk-Sobczyk form factor
 - with Berger-Sehgal lepton mass effects.
 - It accounts for interference among resonances.
 - Nonresonant background is included **incoherently**.
 - **Interference terms are not included.**
- ▶ **Several state-of-the-art SPP models were (are being) implemented.**
 - **ANL-Osaka Dynamical Coupled-Channels (DCC) model**
 - MK model and HNV model ← ongoing



- [1] D. Rein and L. M. Sehgal, *Ann. Phys. (N.Y.)* 133, 79 (1981).
 [2] Graczyk-Sobczyk, *PRD* 77, 053001 (2008).
 [3] C. Berger and L. M. Sehgal, *Phys. Rev. D* 76, 113004 (2007).
 [4] S. X. Nakamura et al., *Phys. Rev. D* 92, 074024 (2015).

- ▶ Initial implementation to NEUT was conducted [5].
- ▶ DCC has larger Xsec than Berger-Sehgal for $\nu_\mu p \rightarrow \mu^- p \pi^+$.
 - The relation varies depending on the channel.
- ▶ Implemented also for electron-pion production [6].

[5] K. Yamauchi et al.,
Proc. Sci. TAUP2023 (2024) 340
[6] S. Abe, PRD 111, 033006 (2025).



Previous

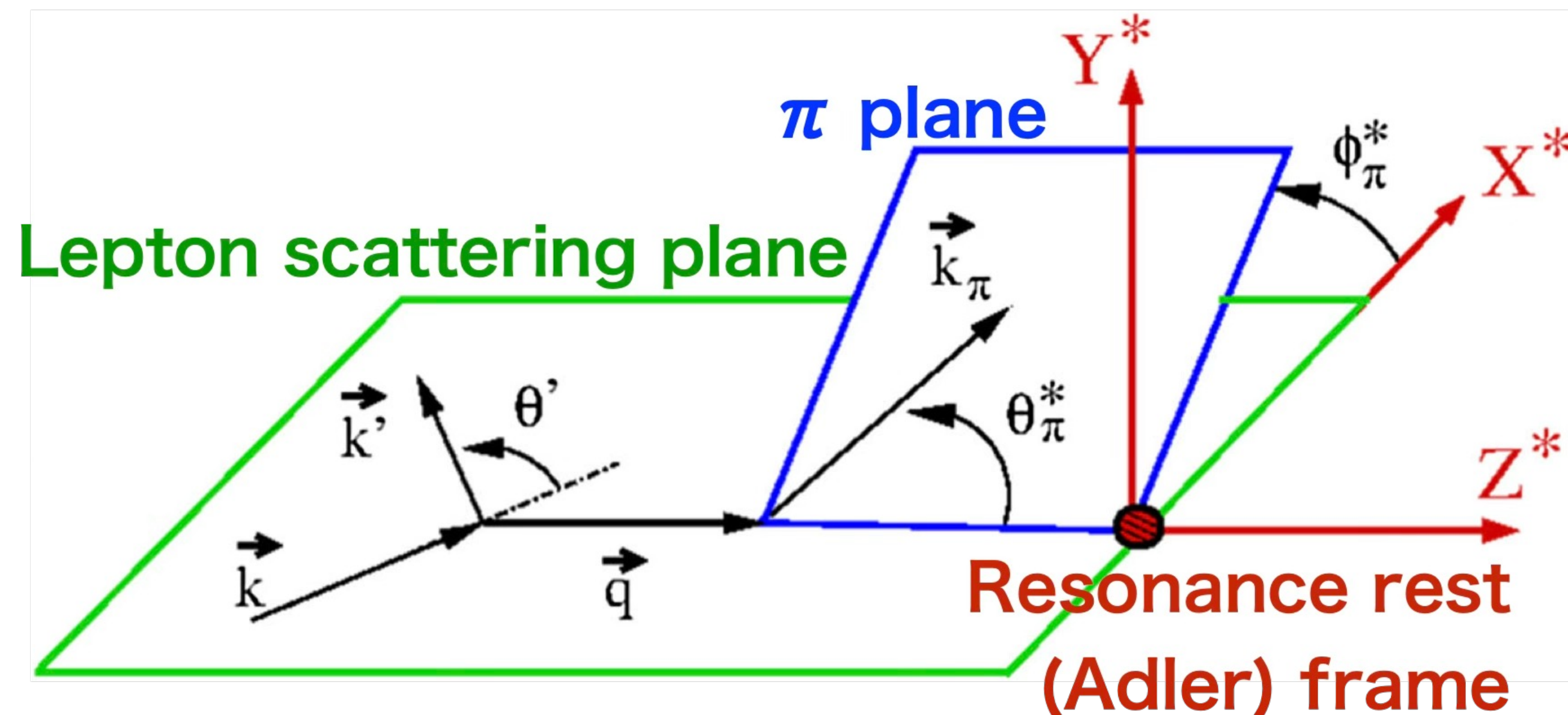
$$\frac{d^2\sigma}{dWdQ^2}(E_\nu)$$

- ▶ Isotropic $\cos \theta_\pi^*$ and ϕ_π^* (Adler angles) distributions are assumed.
- ▶ Tabulated structure functions are used.

New

$$\frac{d^4\sigma}{dWdQ^2 d\cos \theta_\pi^* d\phi_\pi^*}(E_\nu)$$

- ▶ $\cos \theta_\pi^*$ and ϕ_π^* are calculated & sampled as well.
- ▶ Achieved by having the tabulated multiple amplitude.
- ▶ Preliminary results were shown in NuInt2026. Completed implementation after that.

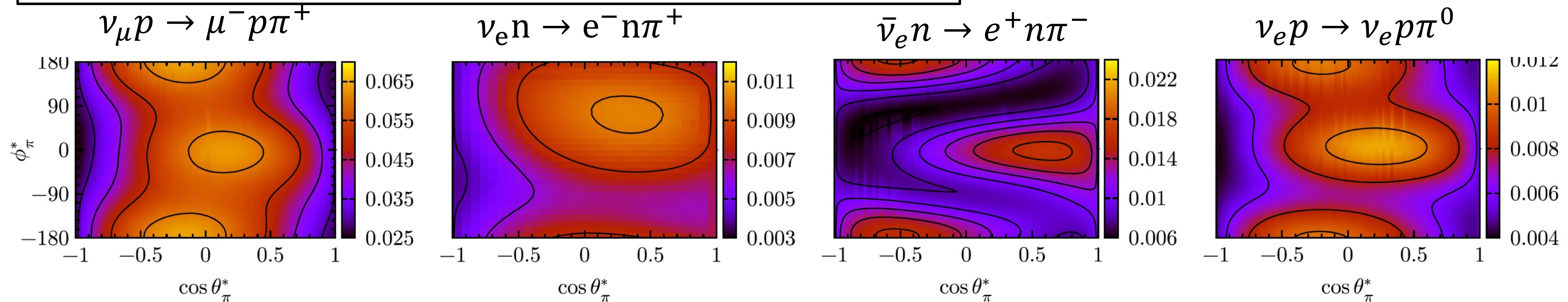


J. E. Sobczyk et al., Phys. Rev. D 98, 073001 (2018).

- ▶ Additional dimensions make event sampling slow.
- ▶ But, after optimization, it's sufficiently fast.
 - ~10 min/100k evts.
 - c.f.) previous: ~2-3 min/100k evts.

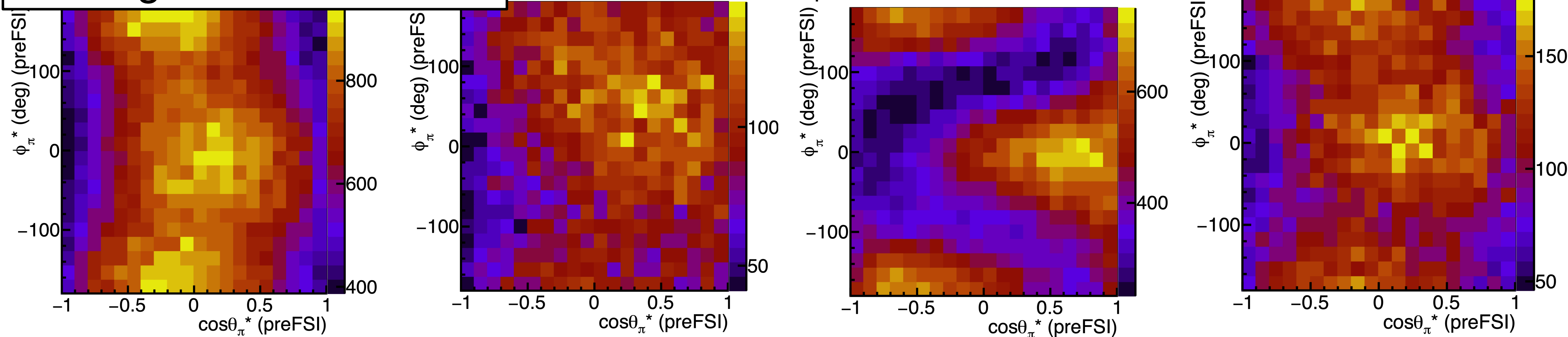
Theoretical calculation: PRD 98, 073001 (2018)

$E_\nu = 1 \text{ GeV}$.



NEUT generated events

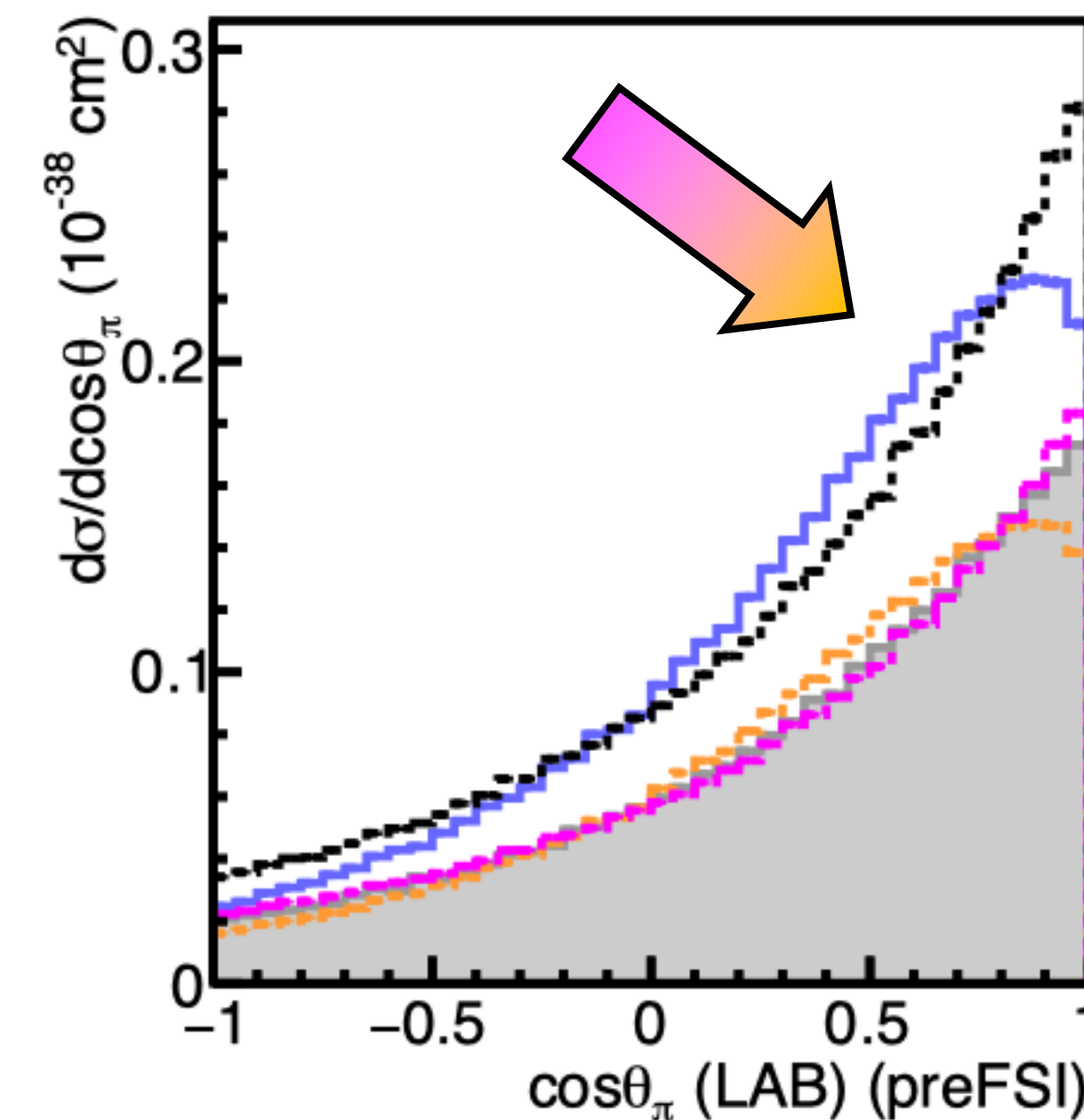
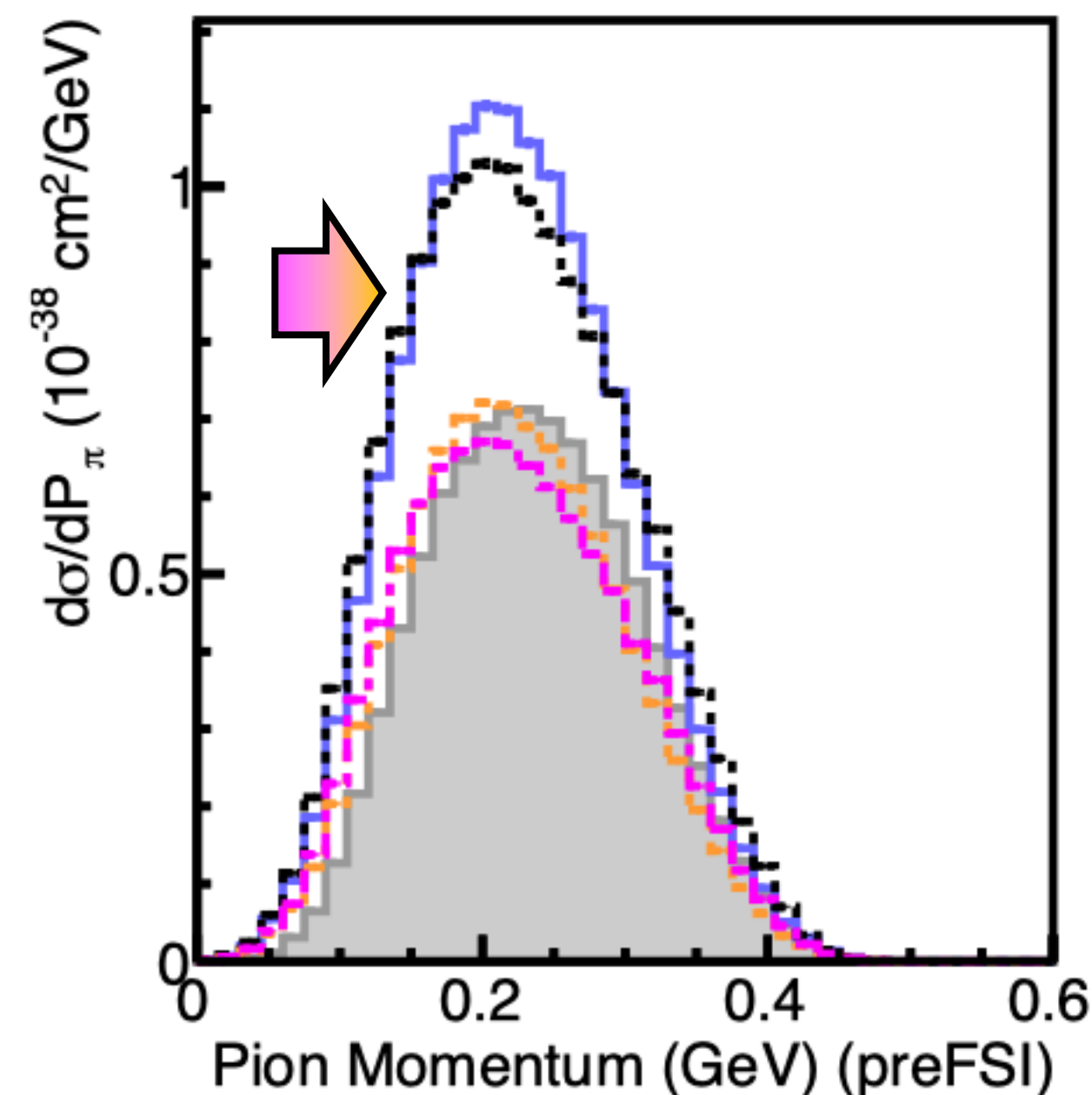
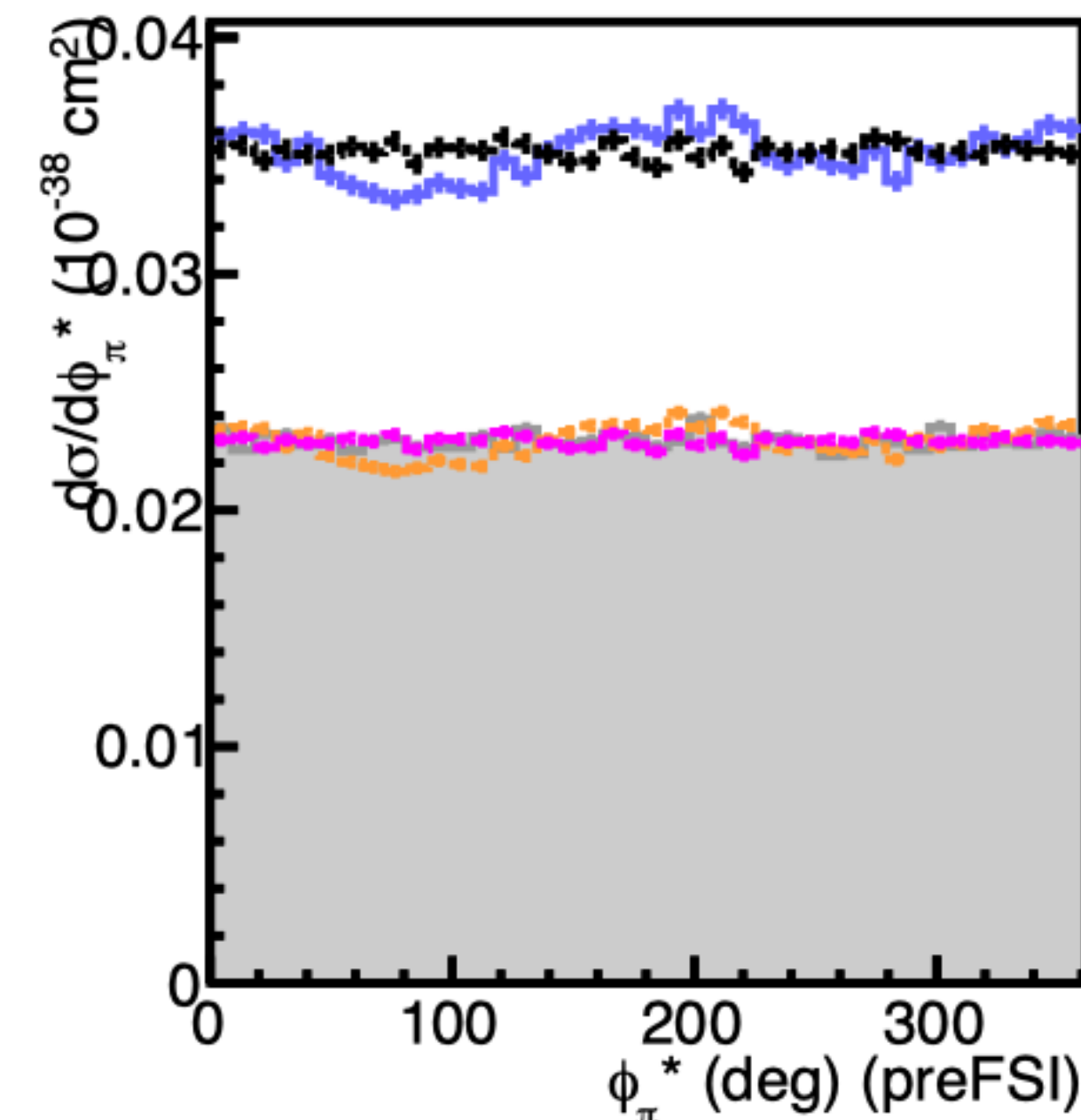
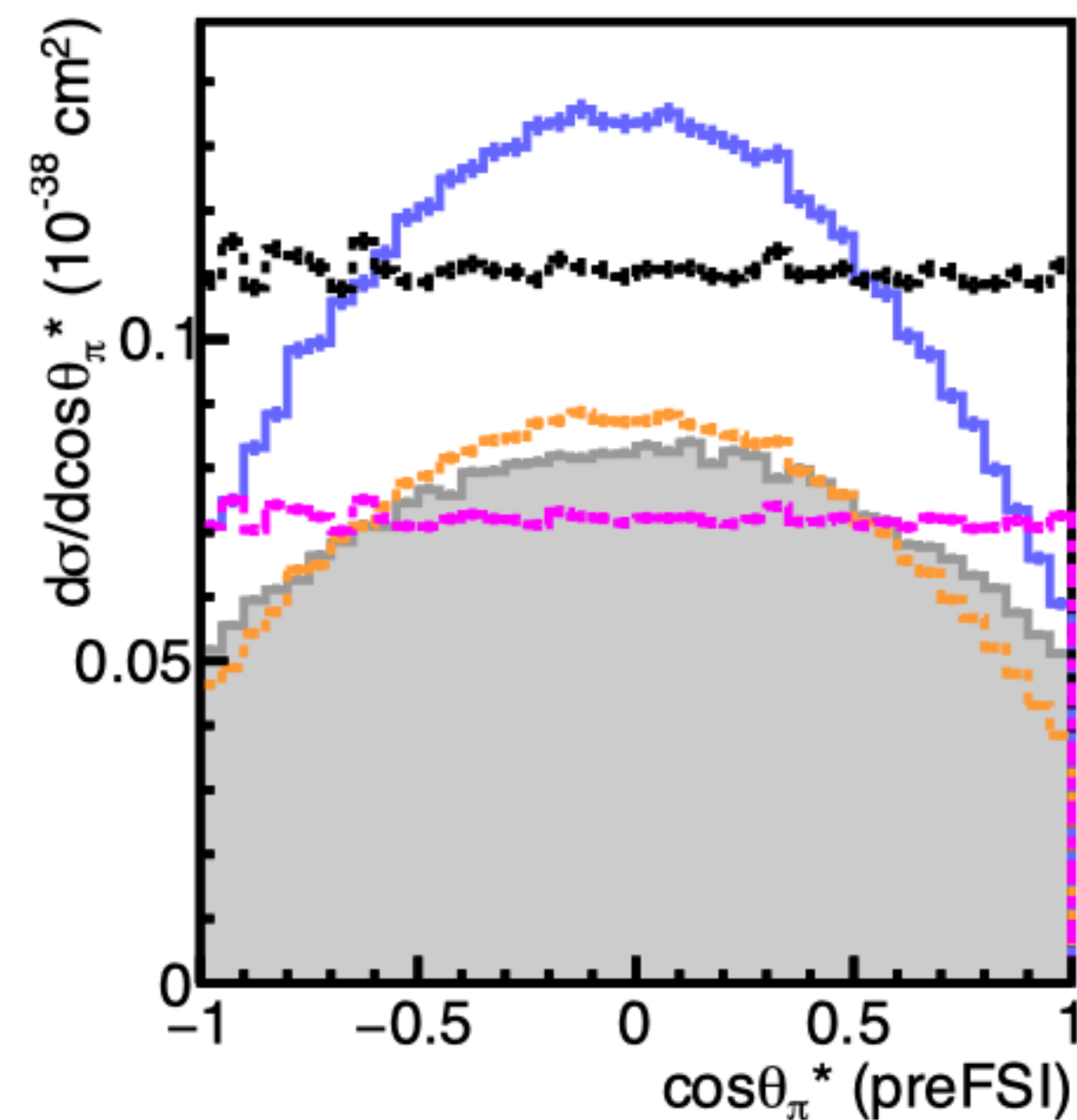
Z-axis : nummbe of events per bin



► **Successful implementation.**

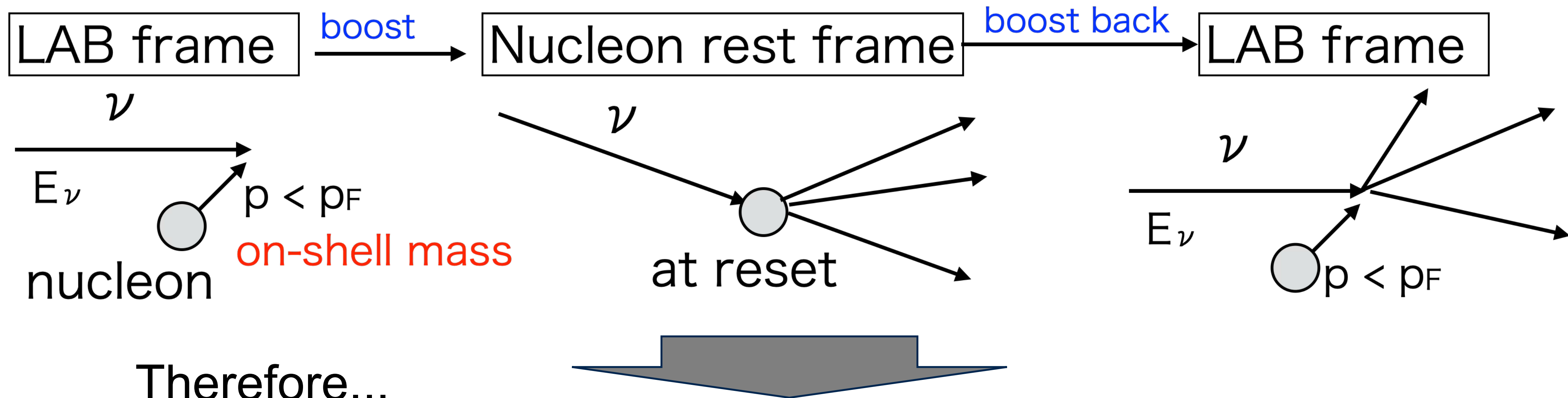
-  Berger-Sehgal (BS)
-  New DCC (aniso)
-  Previous DCC (iso)
-  New DCC (aniso). *Norm to BS.*
-  Previous DCC (iso). *Norm to BS.*

- Inclusion of Adler angle dist.
Distorts pion kinematics.
 - Momentum: slightly larger.
 - Angle: Forward pions are reduced.



Investigation on nuclear model in SPP

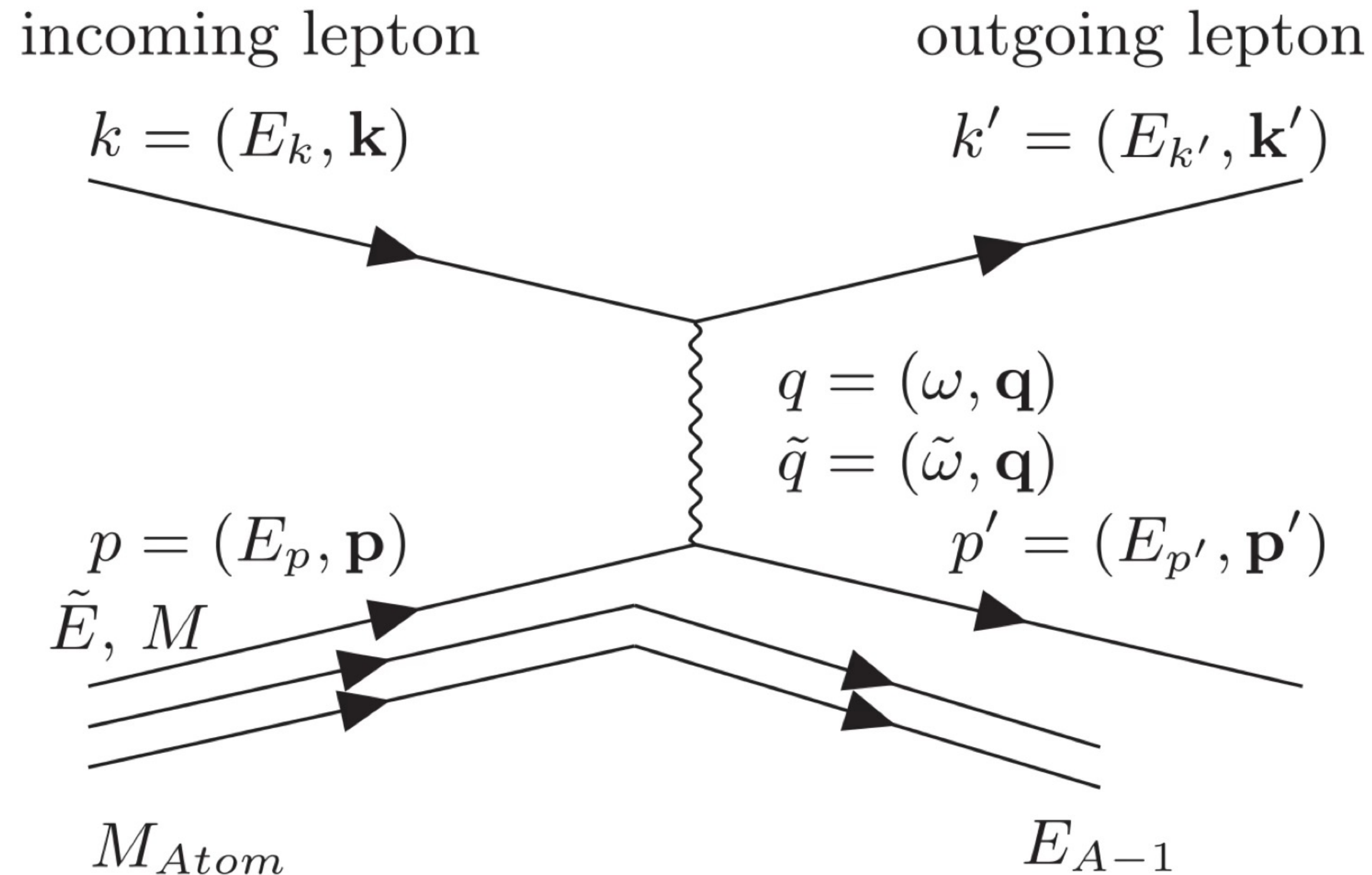
- ▶ All SPP models in NEUT are based on this RFG nuclear model.
- ▶ Assumption:
 - The target nucleon moves below the Fermi surface p_F .
 - The target nucleon has **on-shell mass**.



Therefore...

- ▶ The total energy of outgoing particles exceeds E_ν .
 - It looks as if it has gained energy.
- ▶ Naively expected to be critical in the calorimetric variable 🤔

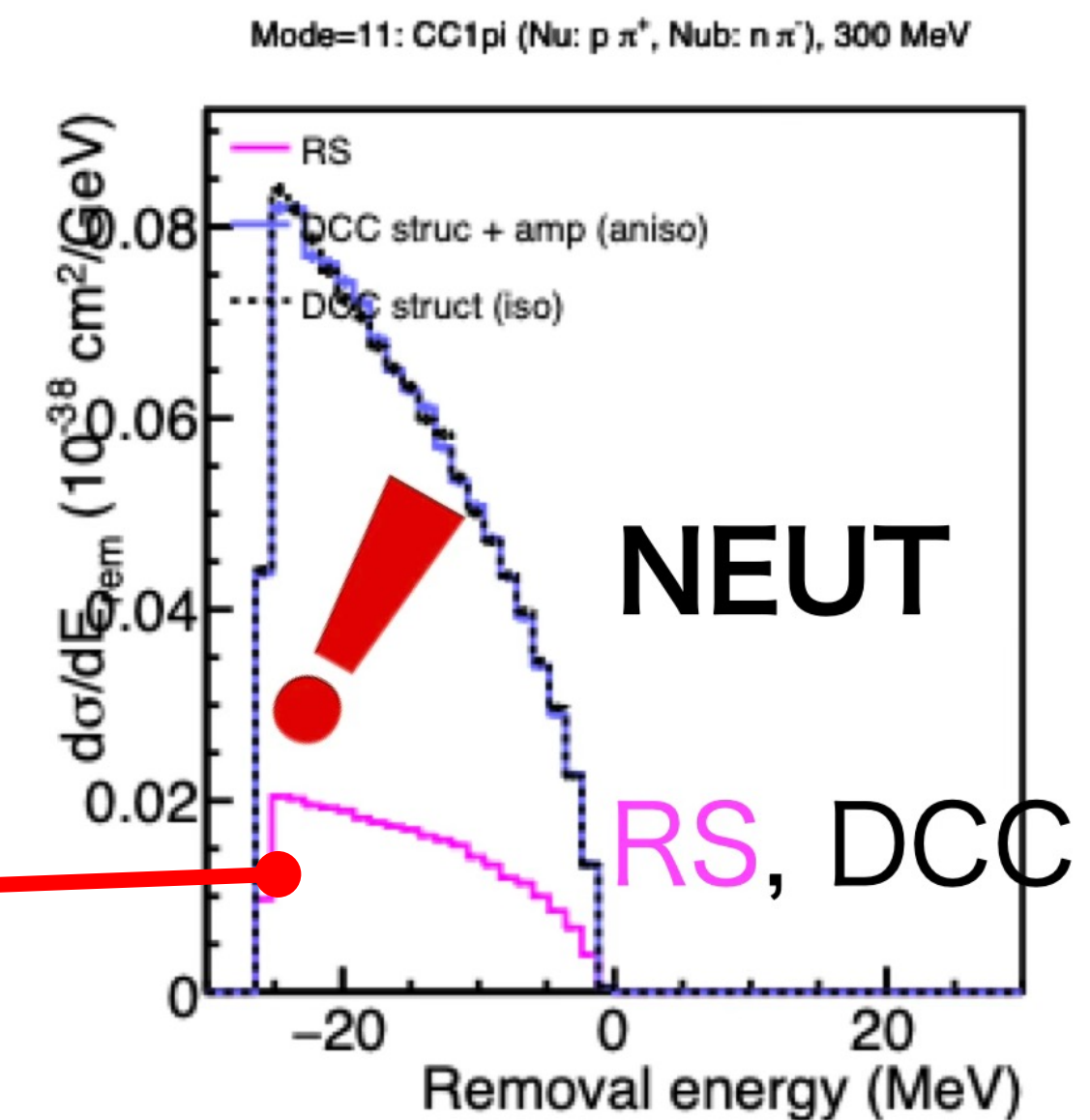
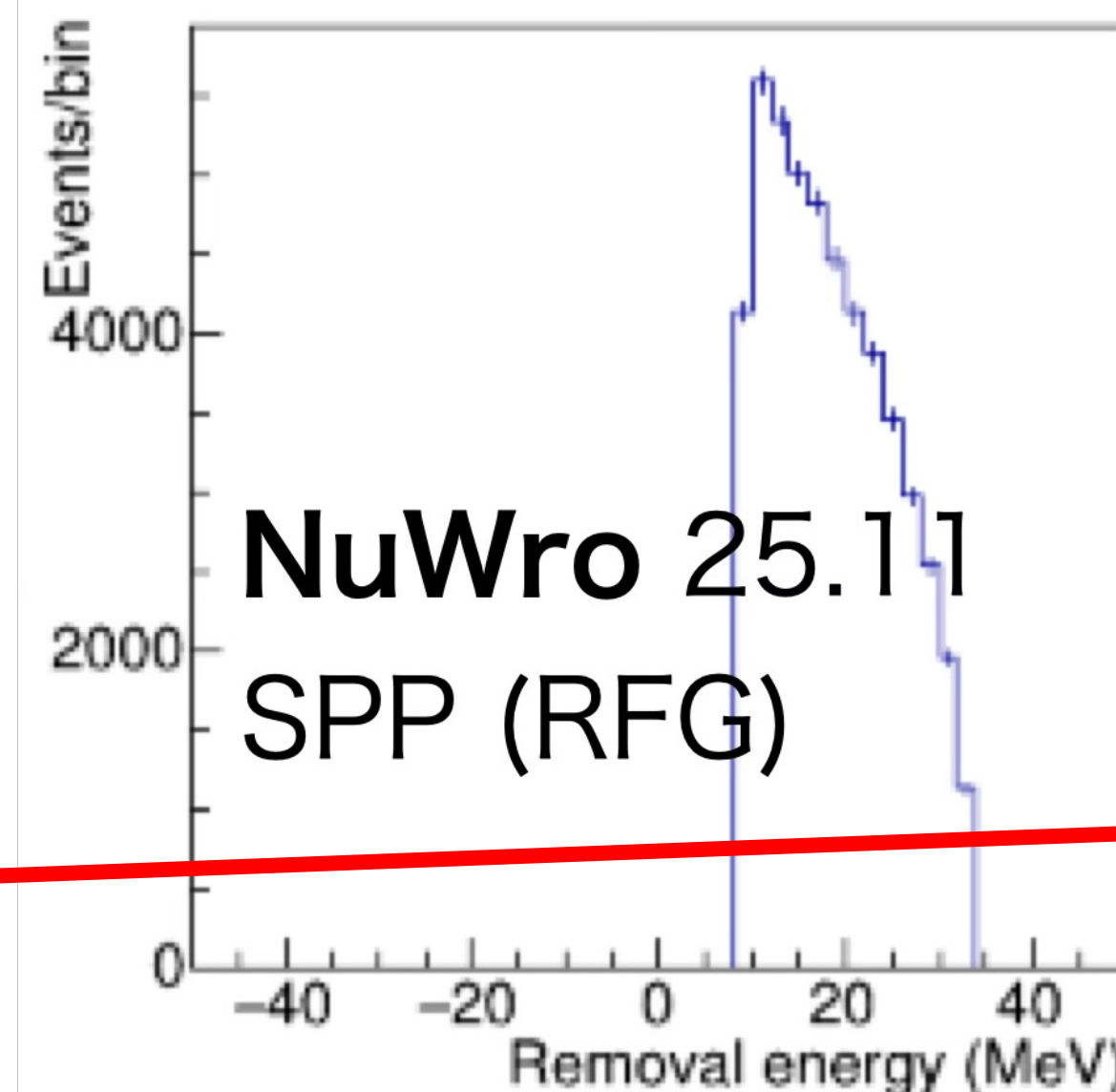
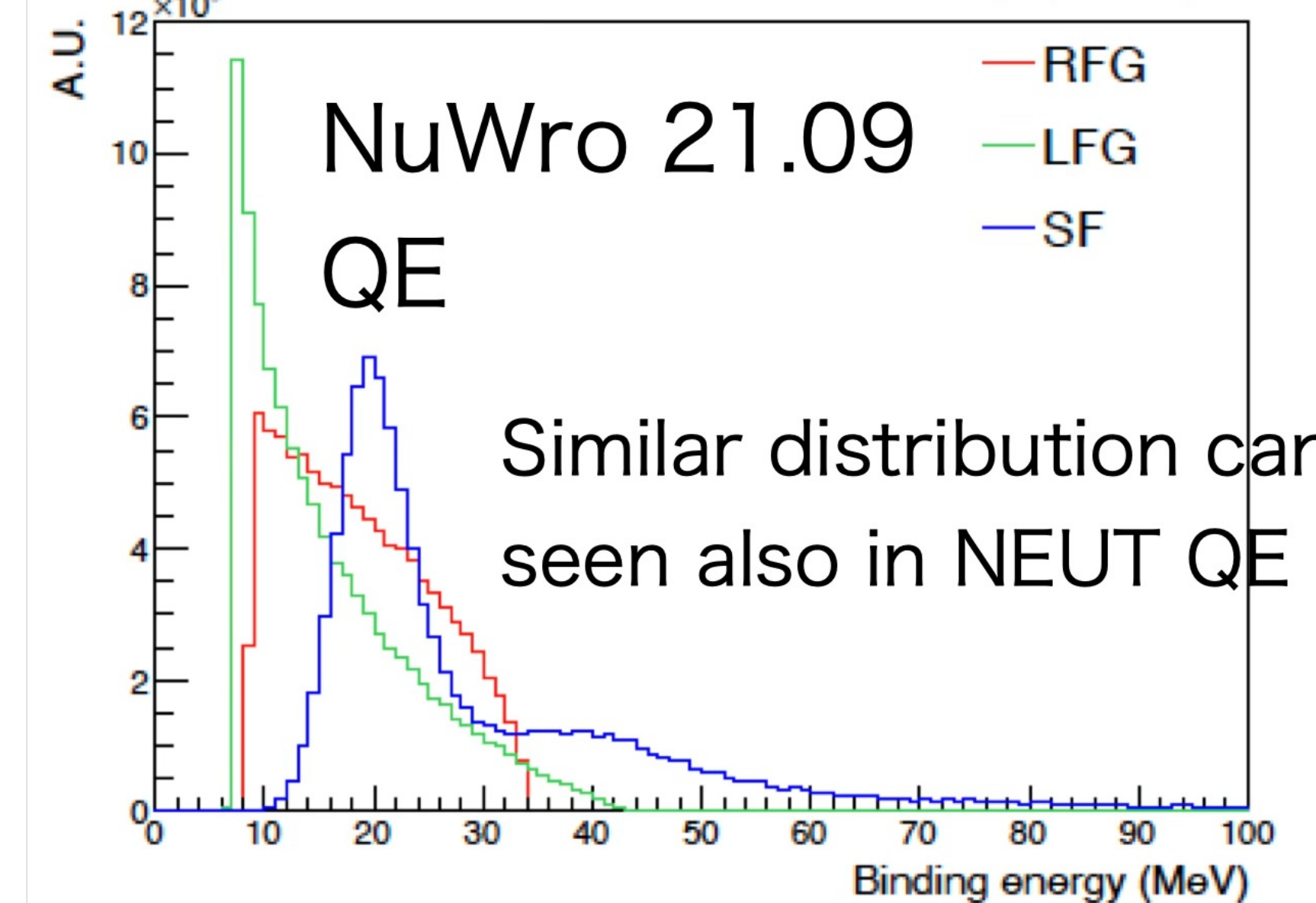
- QE: $E_k + M - E_{k'} - E_{p'}$ neglecting nuclear recoil E.



- SPP: $E_k + M - E_{k'} - E_{p'} - E_{\pi}$

- NEUT SPP gives unphysical & negative removal energies.**

S. Abe, Ph.D. thesis, Tohoku University (2023)



- The same approach as NuWro is taken this time. Implemented to DCC only.
- Introduce binding energy E_b and effective mass M_{eff} .
- The effective mass is used for calculating W and Q^2 .
- It requires **more energy transfer**.

$$p_N = (E, \mathbf{p}) \quad \xrightarrow{\quad} \quad p'_N = (E', \mathbf{p})$$

$$E = \sqrt{p^2 + M^2} \quad \text{on-shell mass} \quad \xrightarrow{\quad} \quad E' = E - E_b$$

Binding energy E_b

$$E_b = \begin{cases} 34 \text{ MeV} & \text{RFG} \\ E_{Fermi} + 7 \text{ MeV} & \text{LFG} \\ \mathbf{E - M} + \tilde{E} & \text{SF} \end{cases}$$

Kinematic energy + Removal E

$$E_{Fermi} = \sqrt{\mathbf{p}_{Fermi}^2 + M^2} - M$$

p_{Fermi} : Radius dependent Fermi momentum.

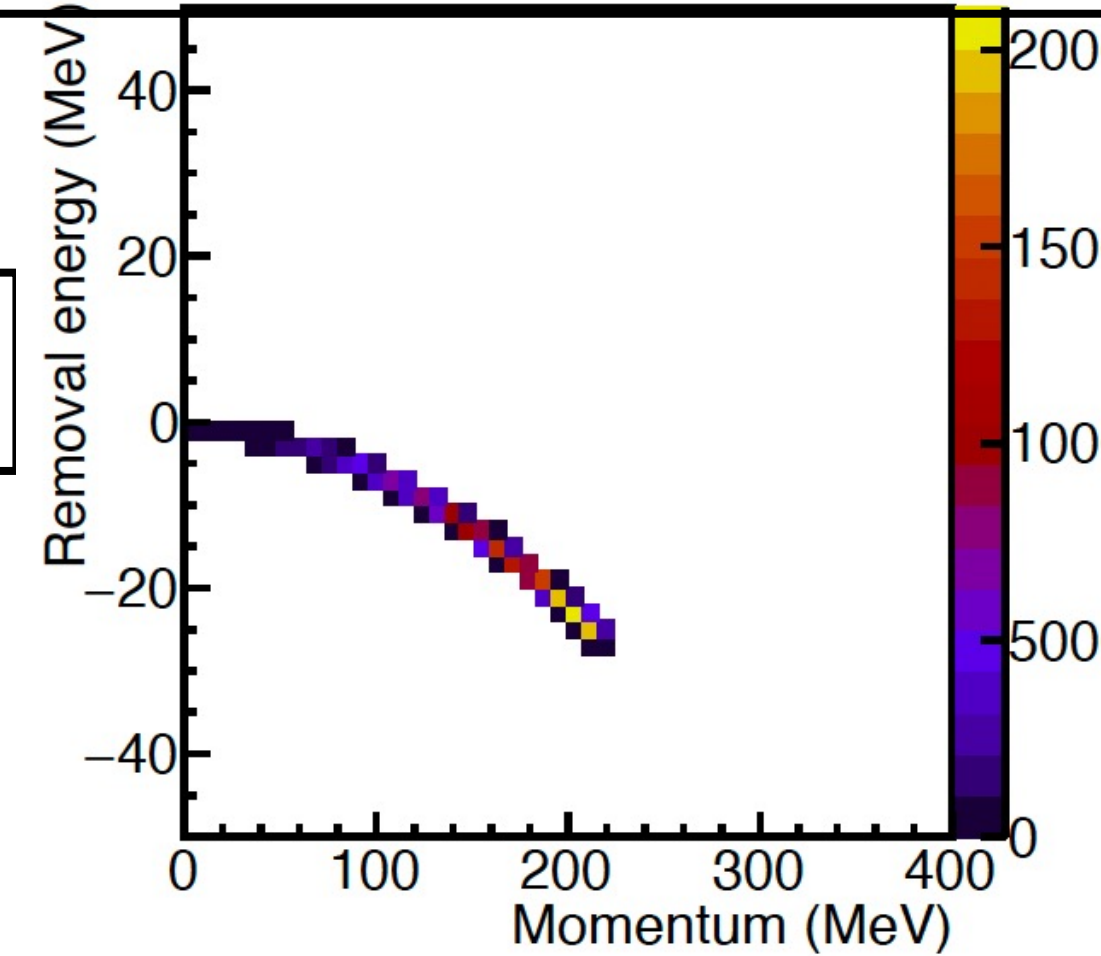
$\tilde{E}$: Removal energy given by SF.

$$M_{eff} = \sqrt{E'^2 - p^2} = \sqrt{(E - E_b)^2 - p^2} \sim M - E_b.$$

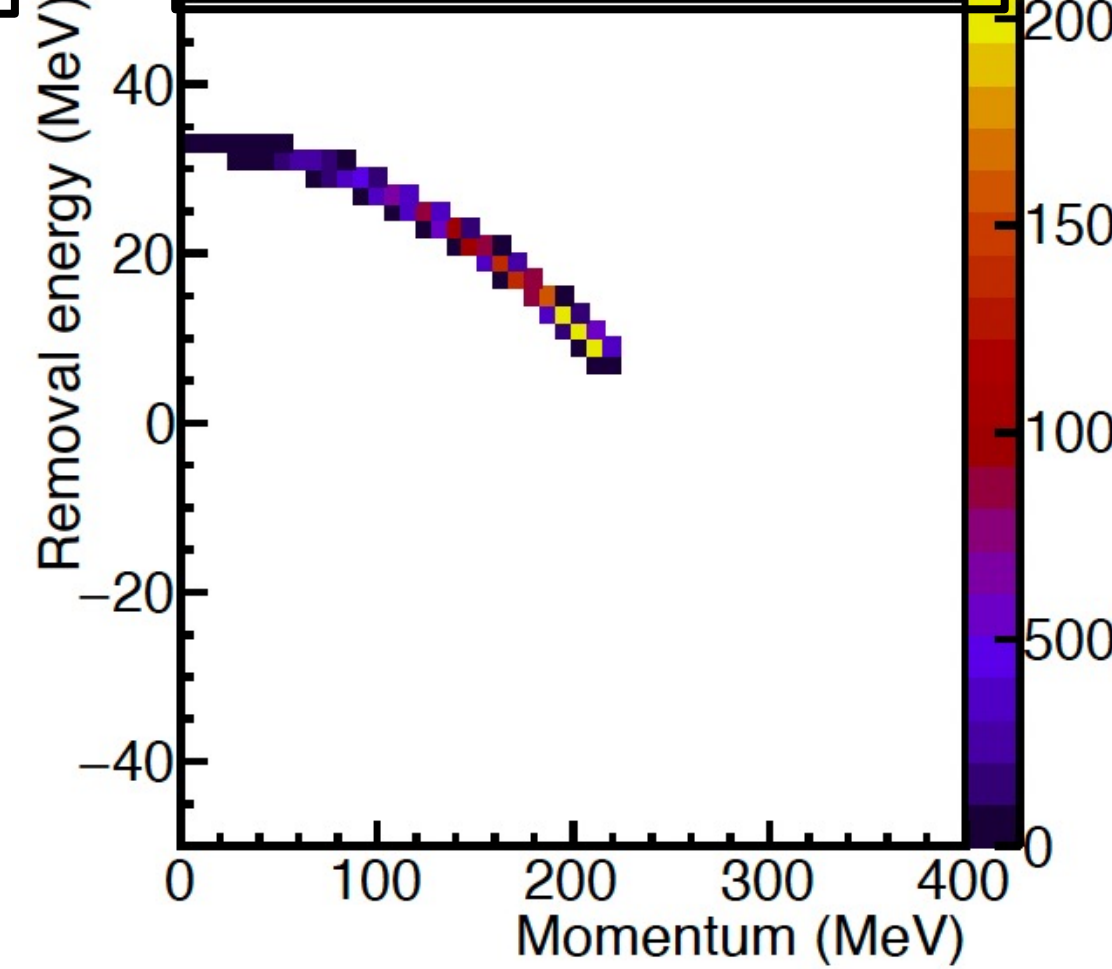
Target nucleon profile of new SPP

- Removal energy for SPP: $E_k + M - E_{k'} - E_{p'} - E_\pi$.

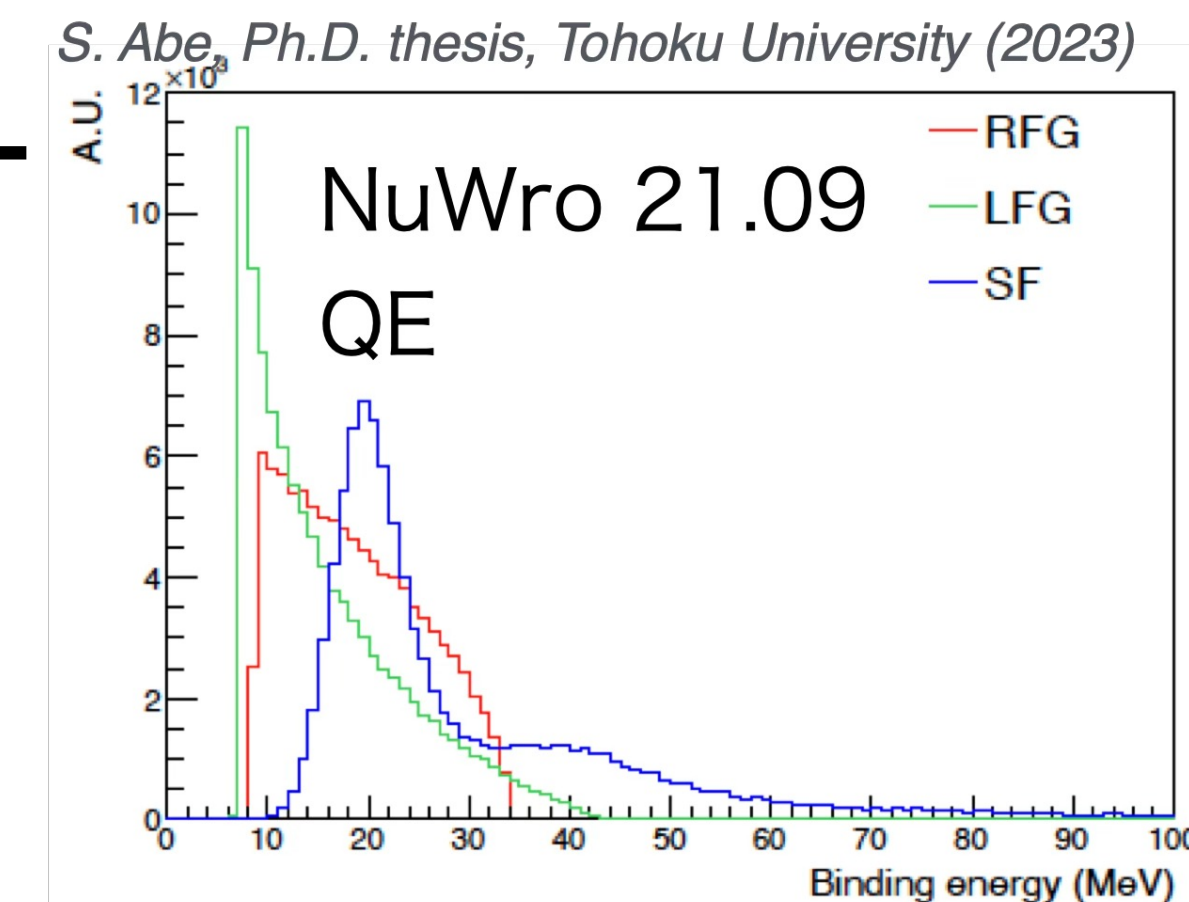
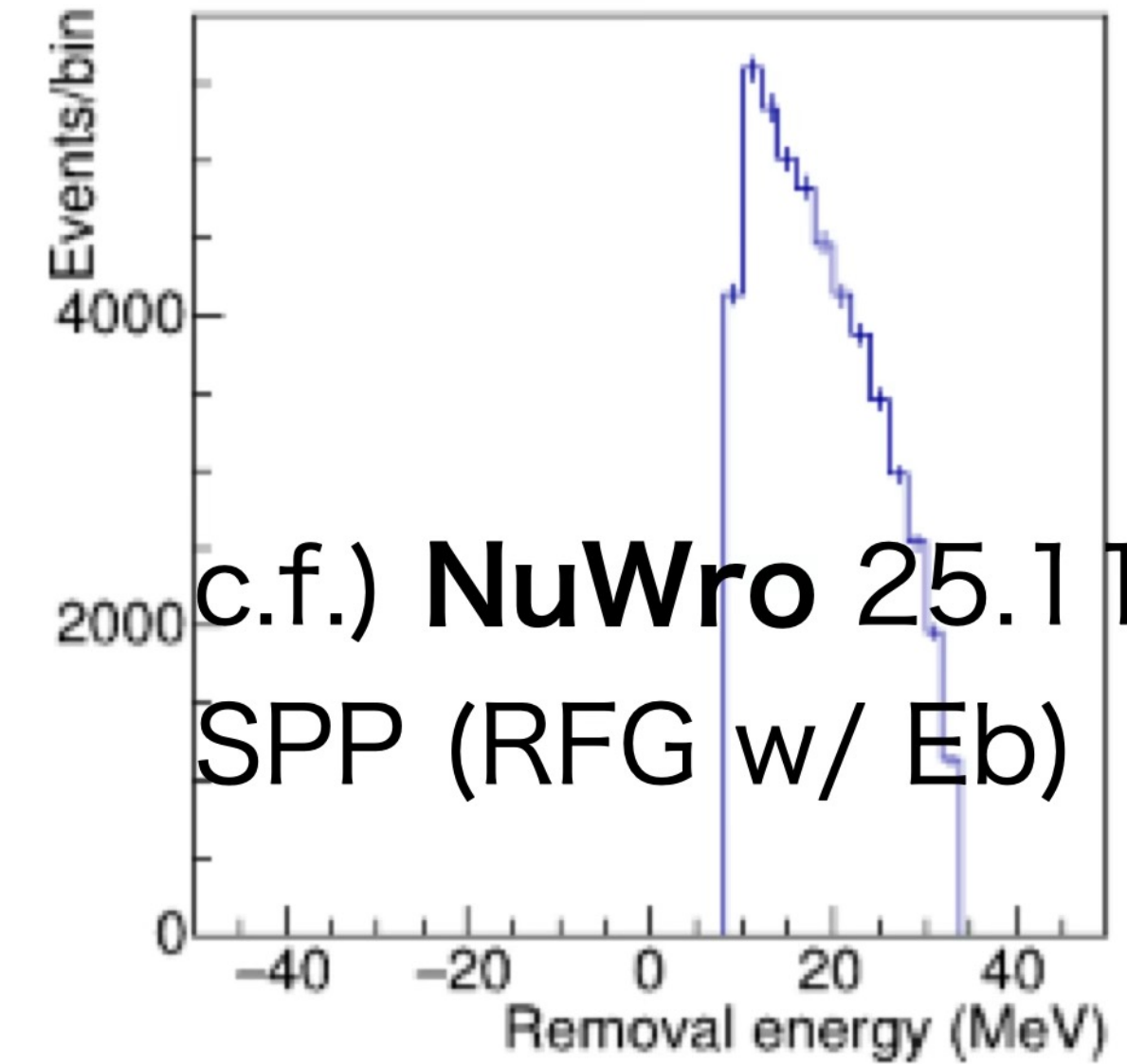
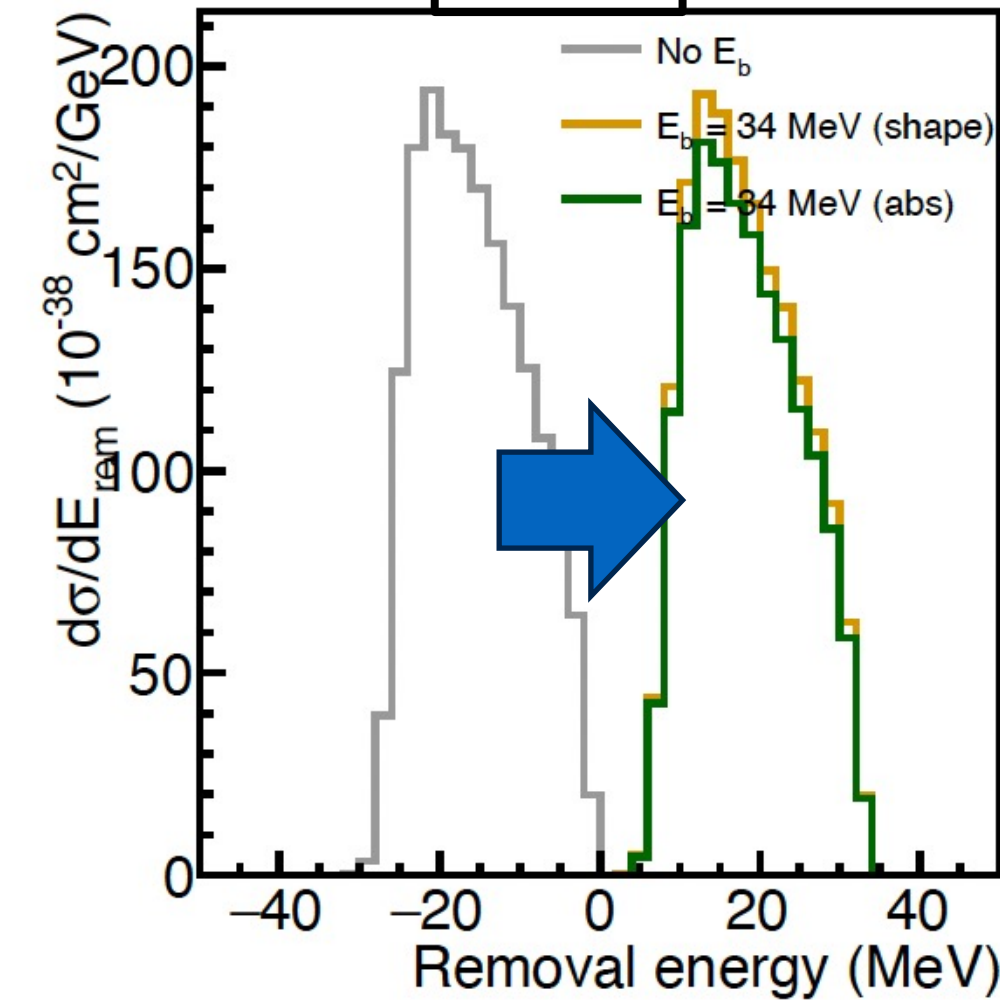
Previous (w/o E_b)



New (w/ E_b)

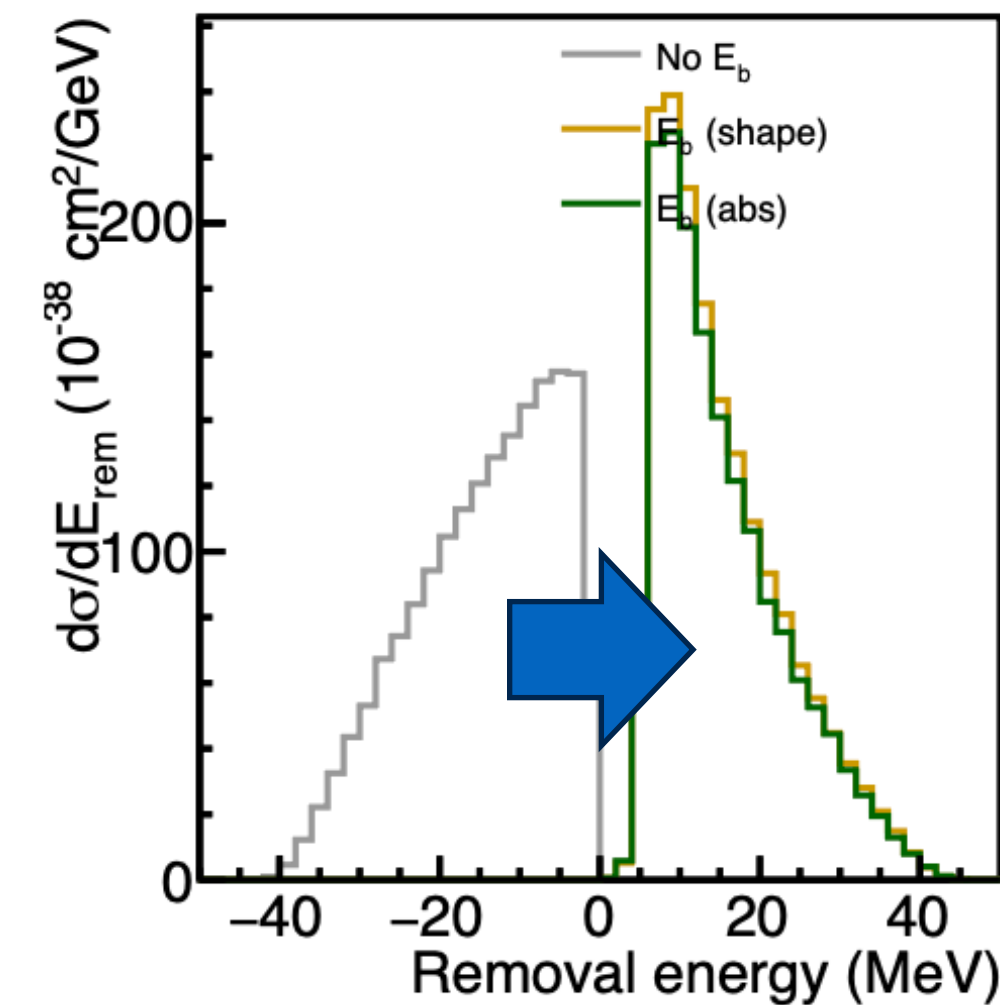
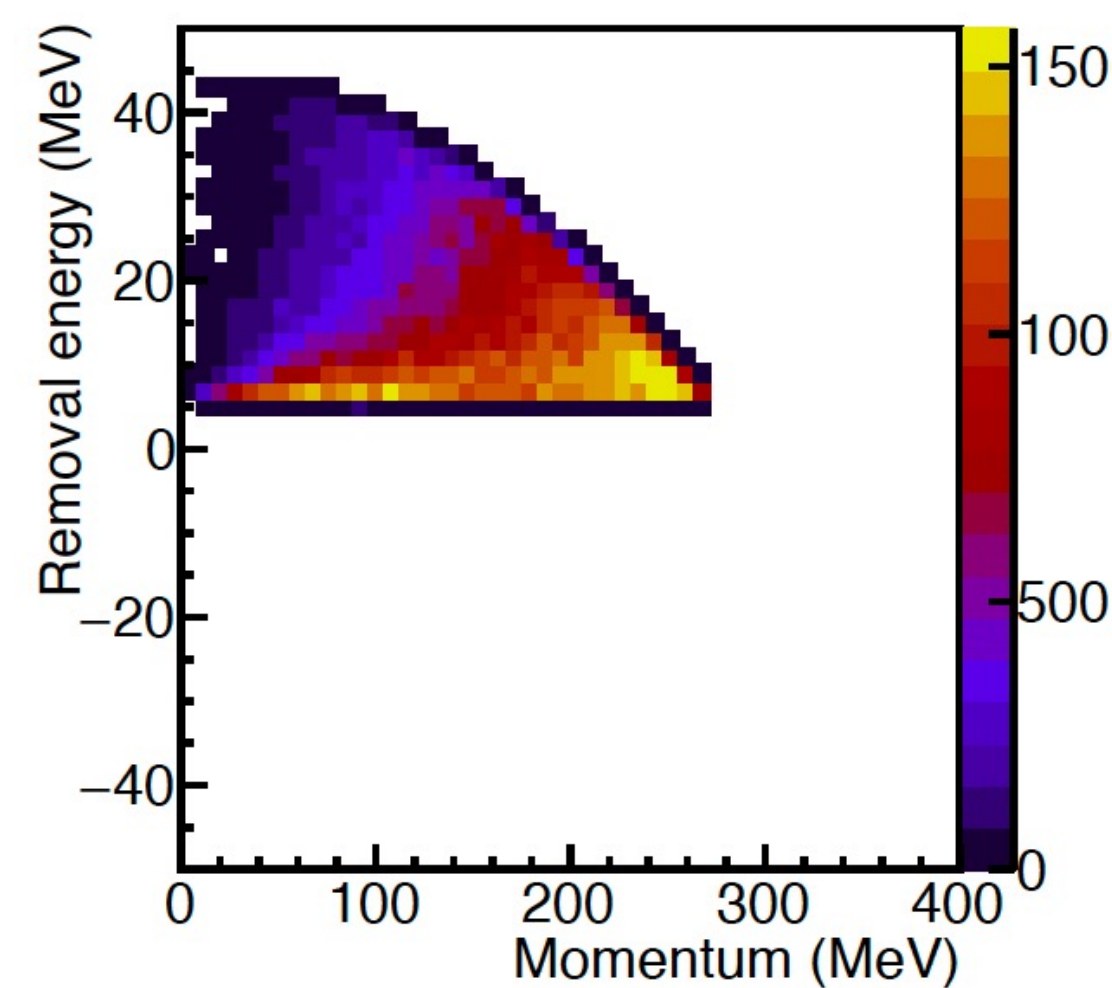
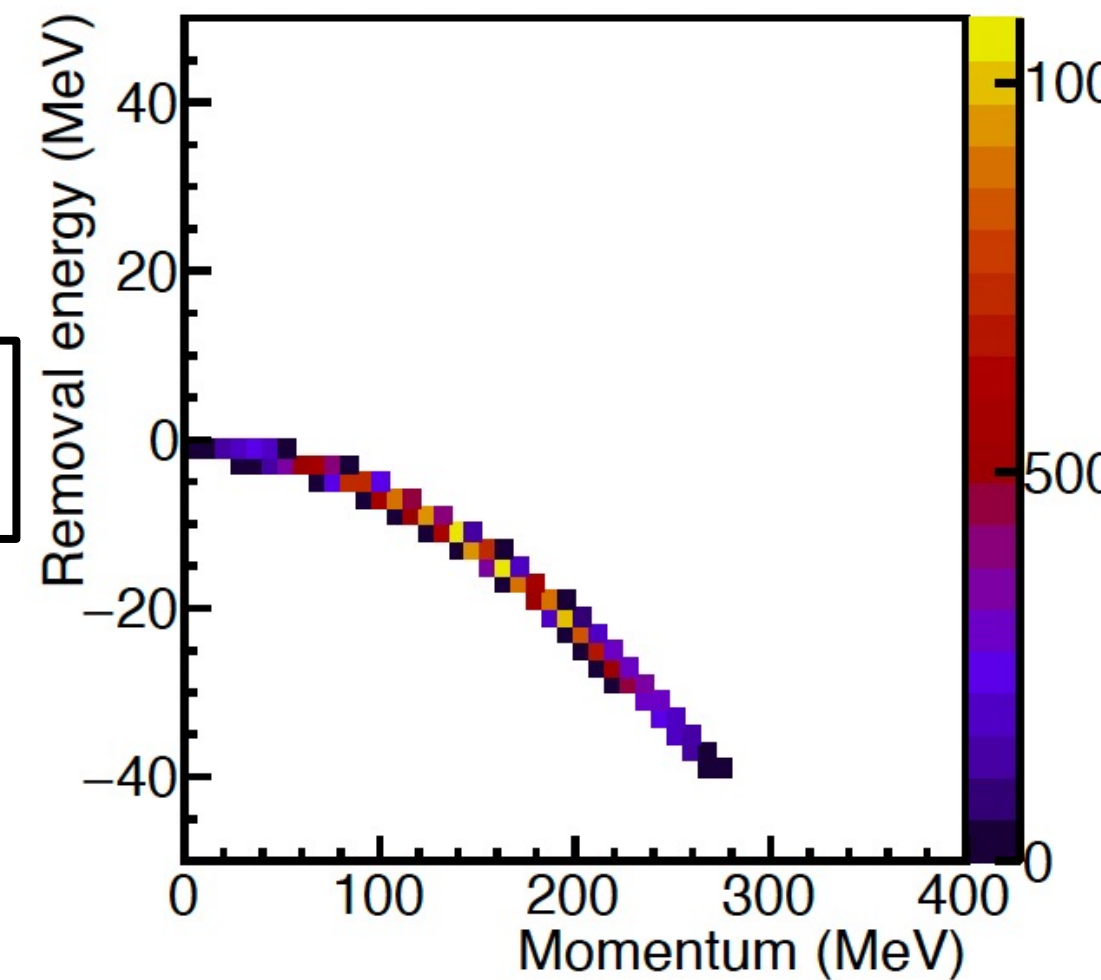


1D



RFG

LFG



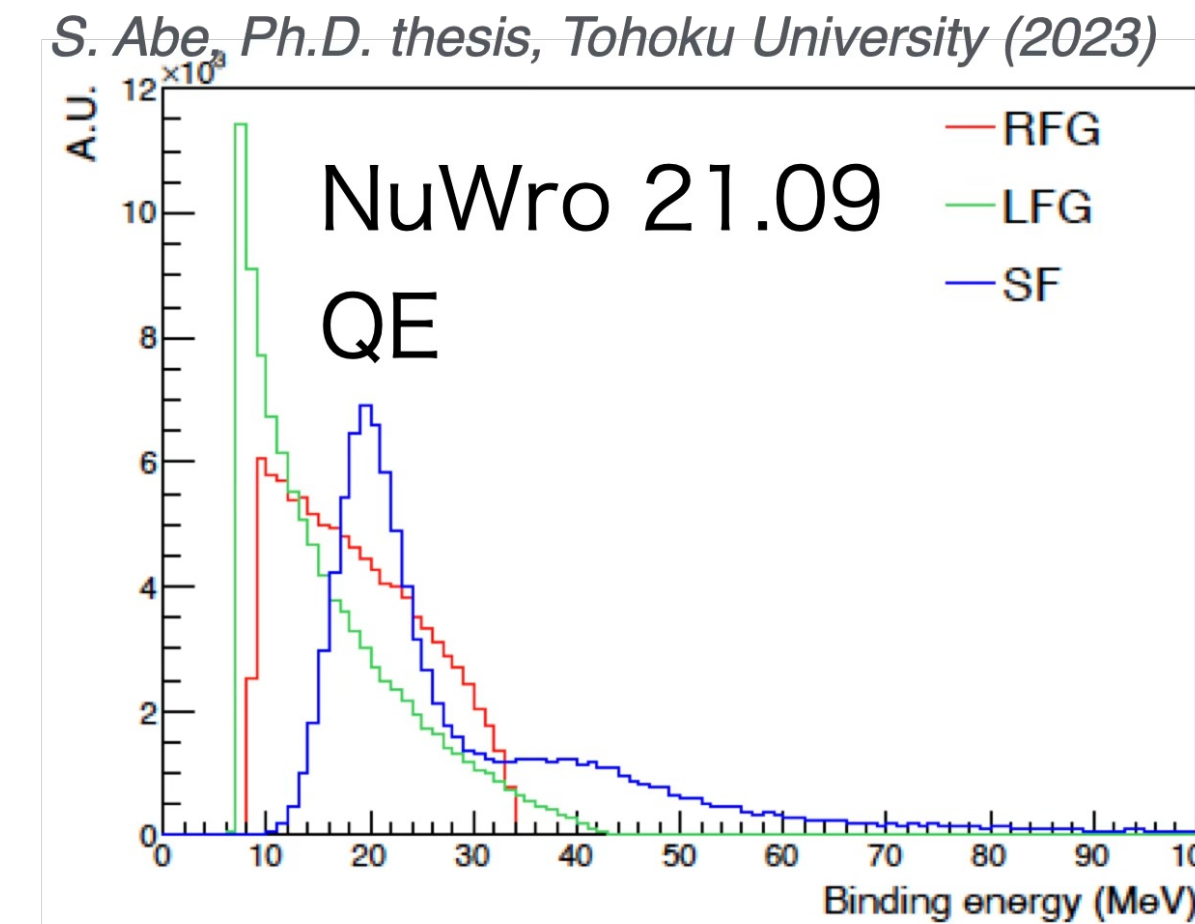
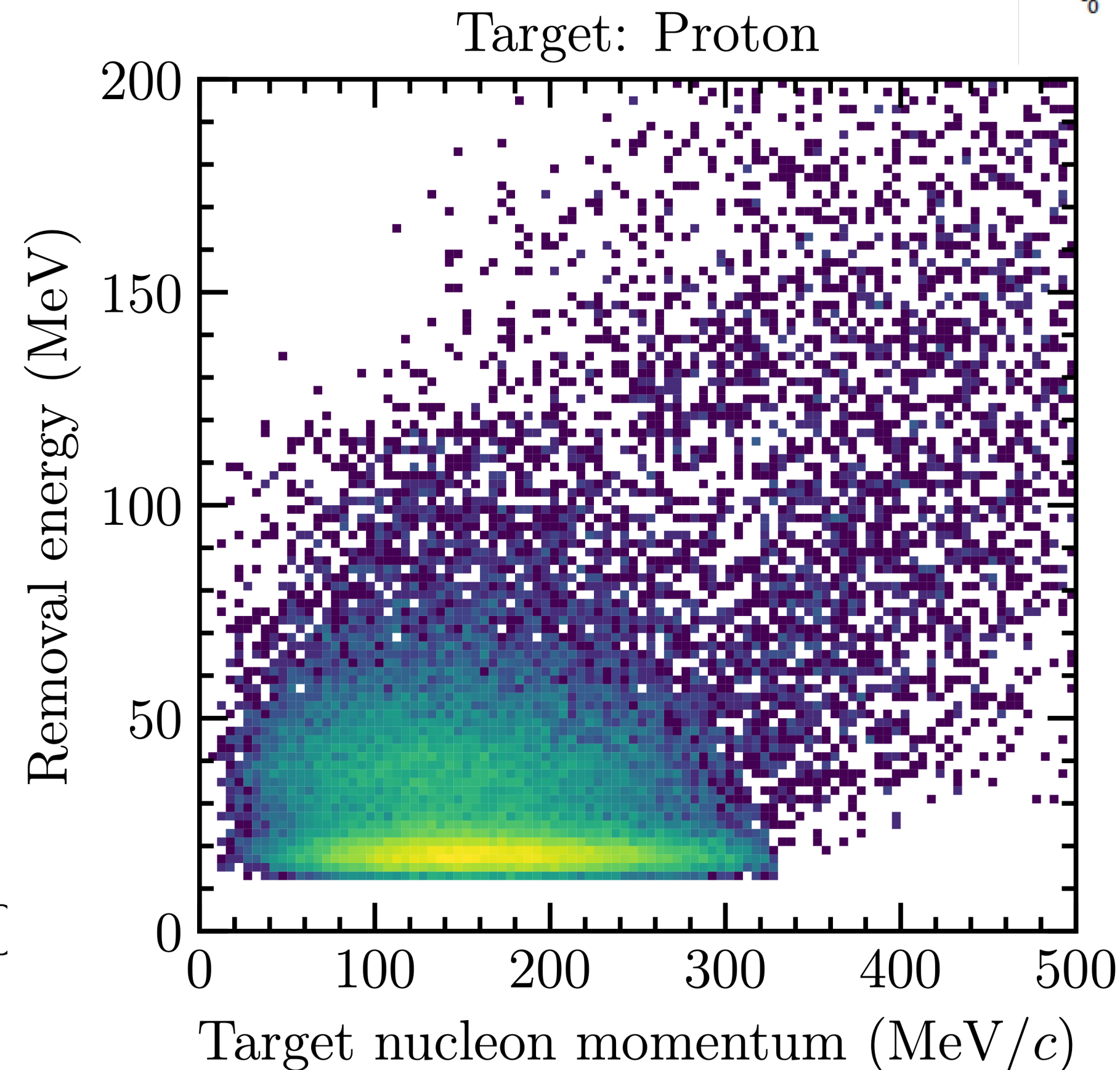
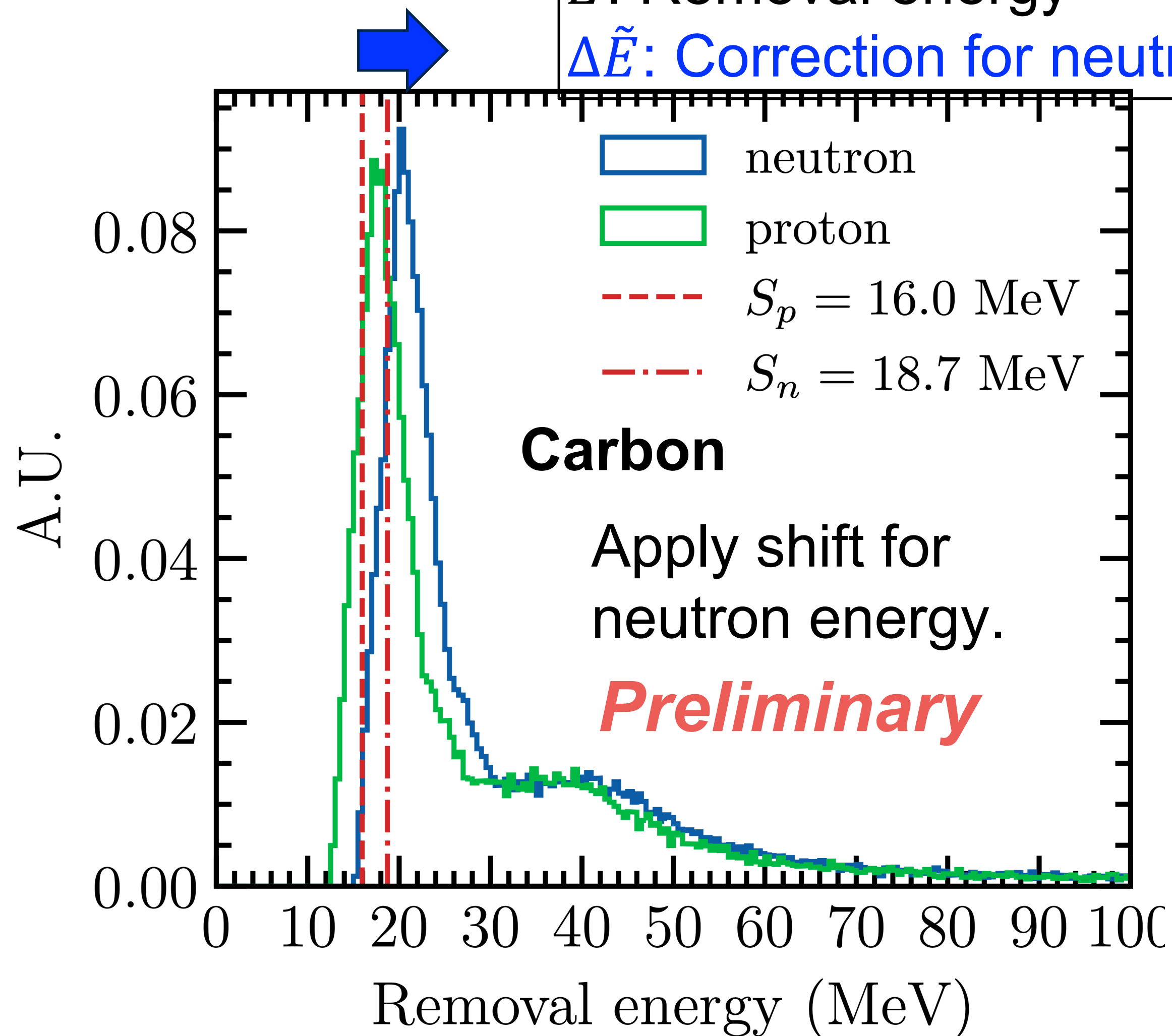
- Well known distributions → Successful implementation.

- **Spectral Function (SF) is implemented.**

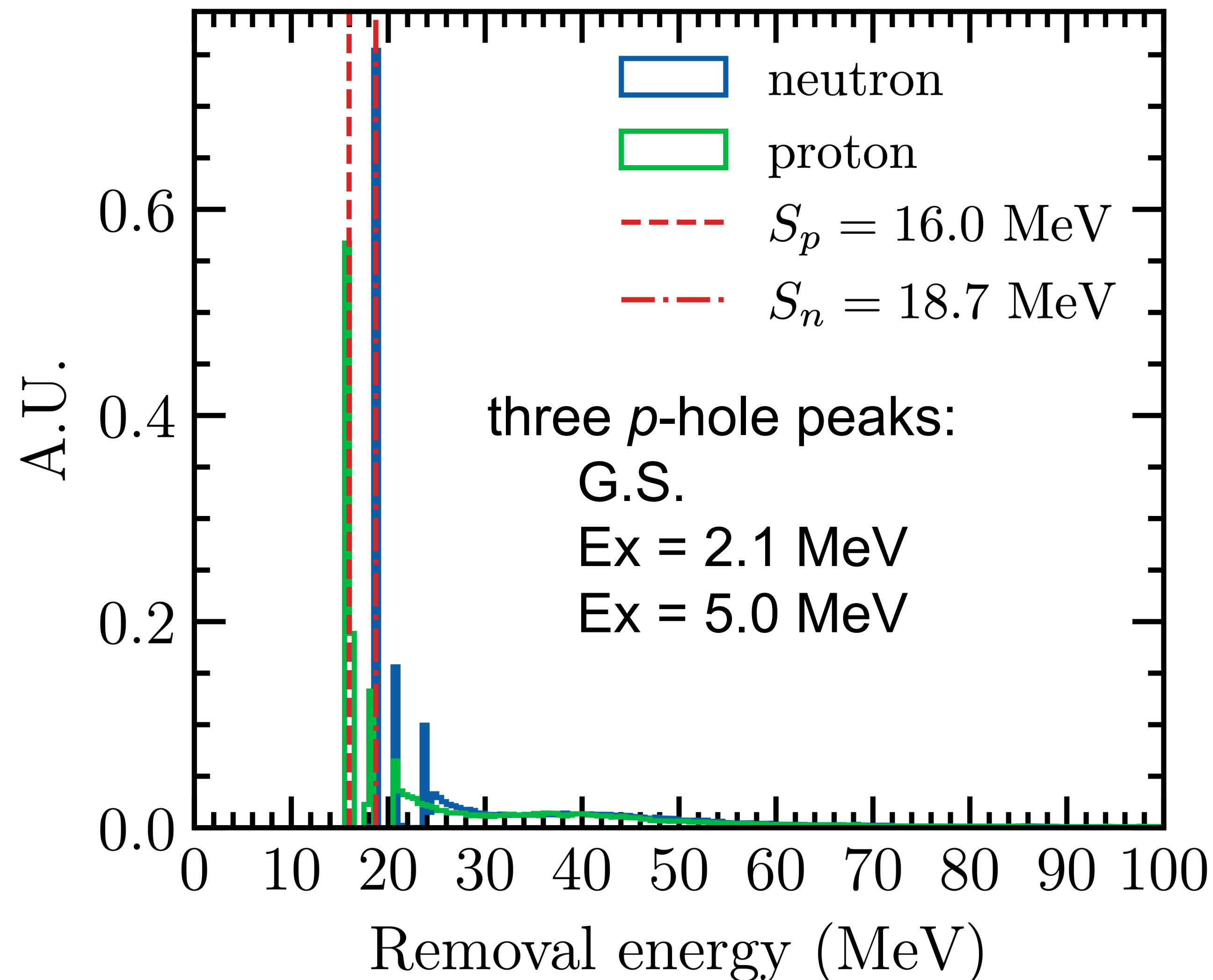
$$P_{\text{neutron}}(\mathbf{p}, \tilde{E}) = P_{\text{proton}}(\mathbf{p}, \tilde{E} - \Delta\tilde{E}),$$

$\tilde{E}$: Removal energy

$\Delta\tilde{E}$: Correction for neutron. 2.76 MeV for carbon.



- ▶ **New carbon SF is also implemented.**
- ▶ Event sampling is slower than previous SF (the same as QE).

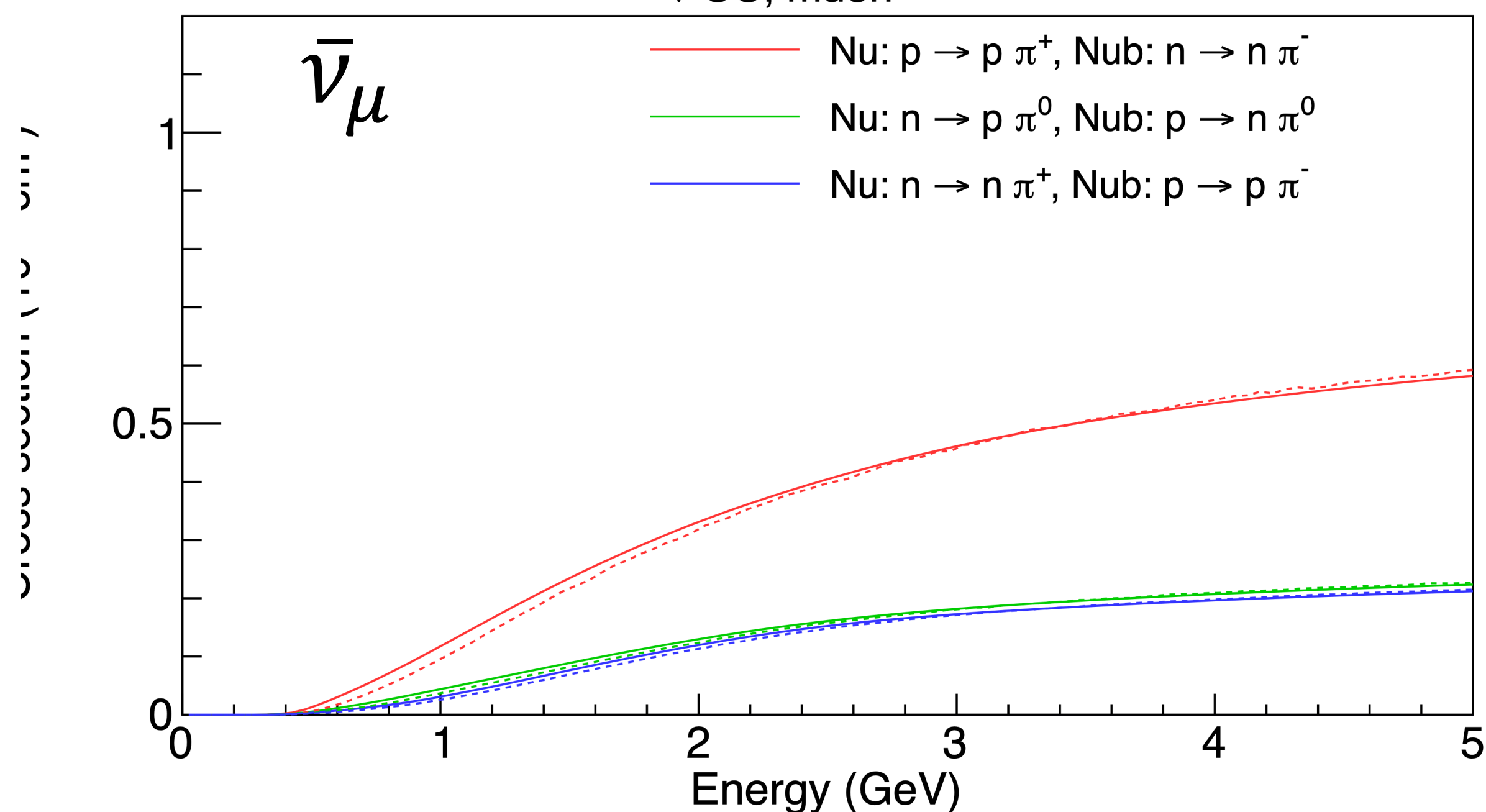
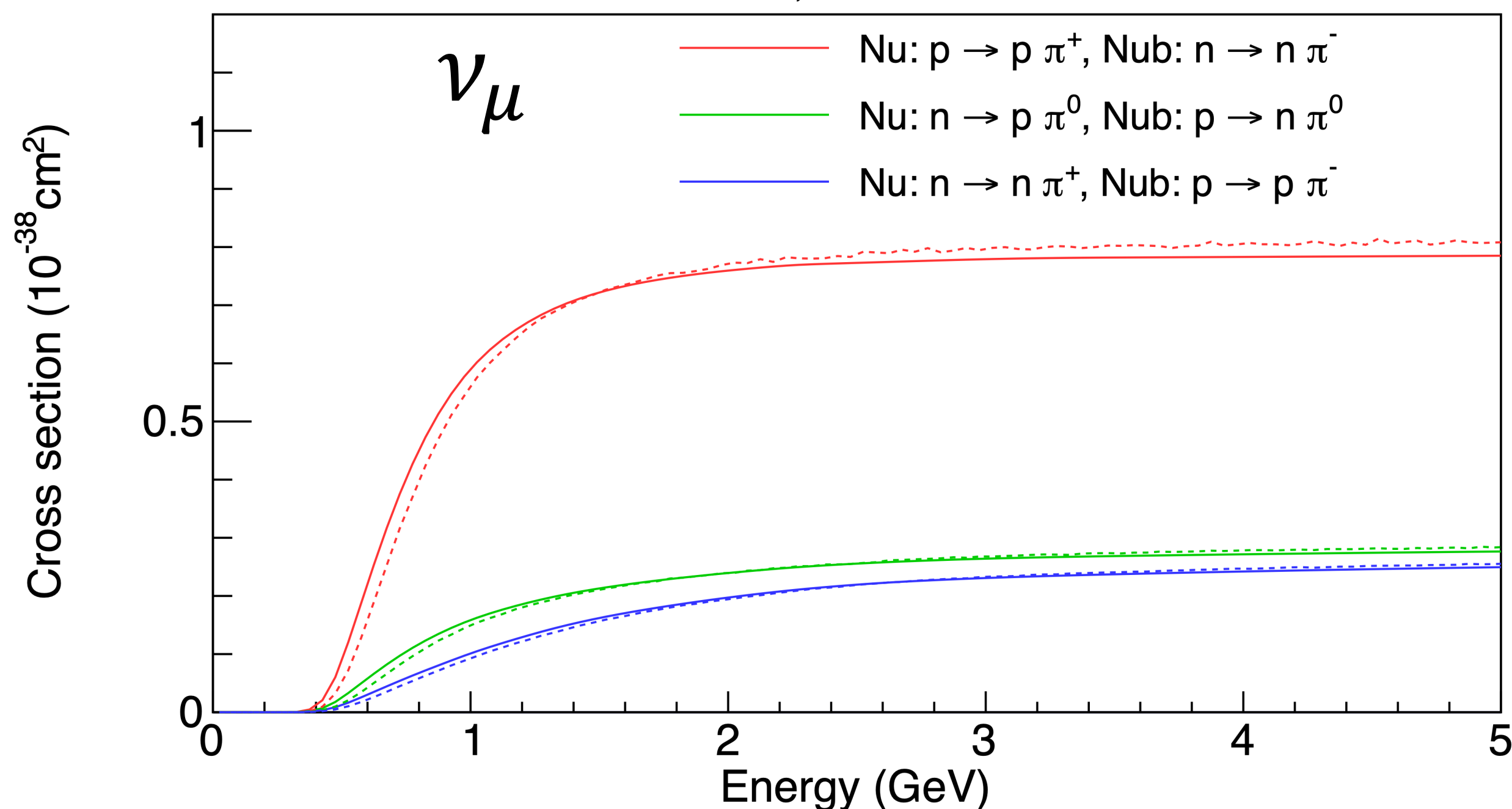
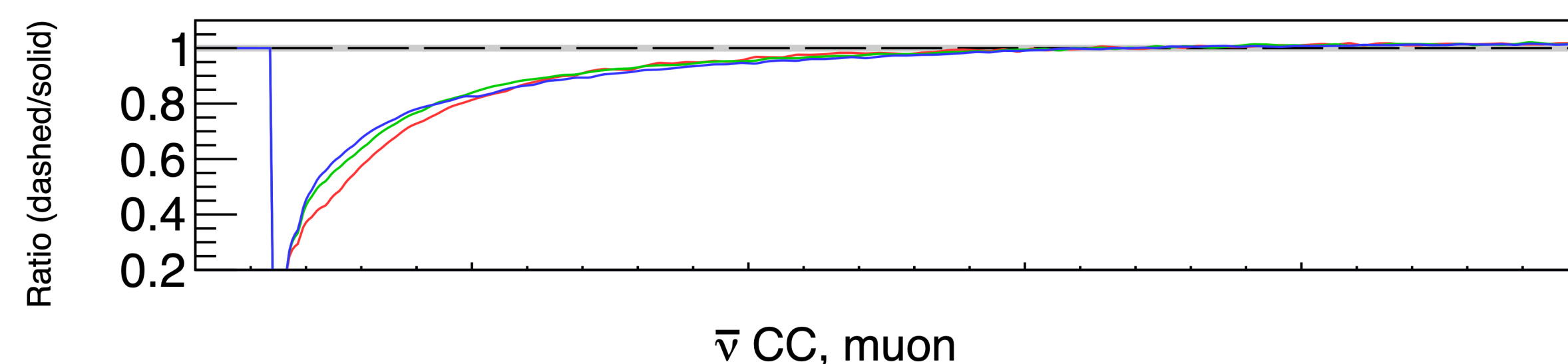
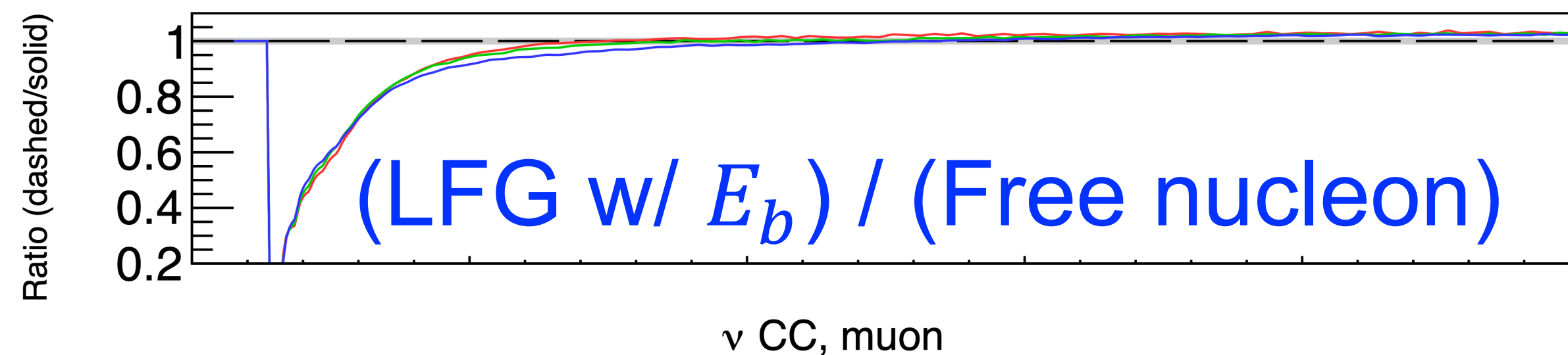


- ▶ **The G.S. peaks appear at the true energy (S_p and S_n).**
- ▶ Note) The new SF does not change lepton kinematics so much.
 - It's important in deexcitation, but not so much for others.

- The total Xsec is suppressed by the E_b at low-E ν .

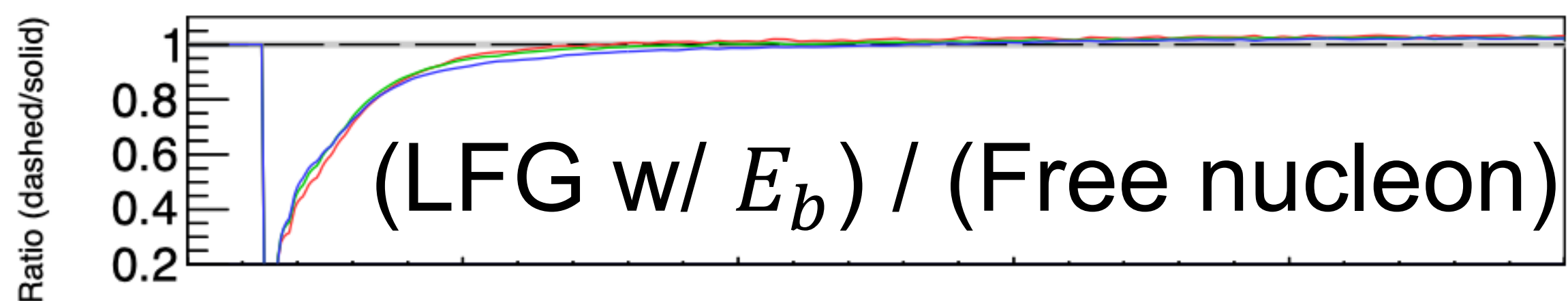
Dashed: LFG w/ E_b
Solid: Free nucleon.

Xsec of LFG is ~60-70% of free nucleon at $E\nu=600$ MeV.

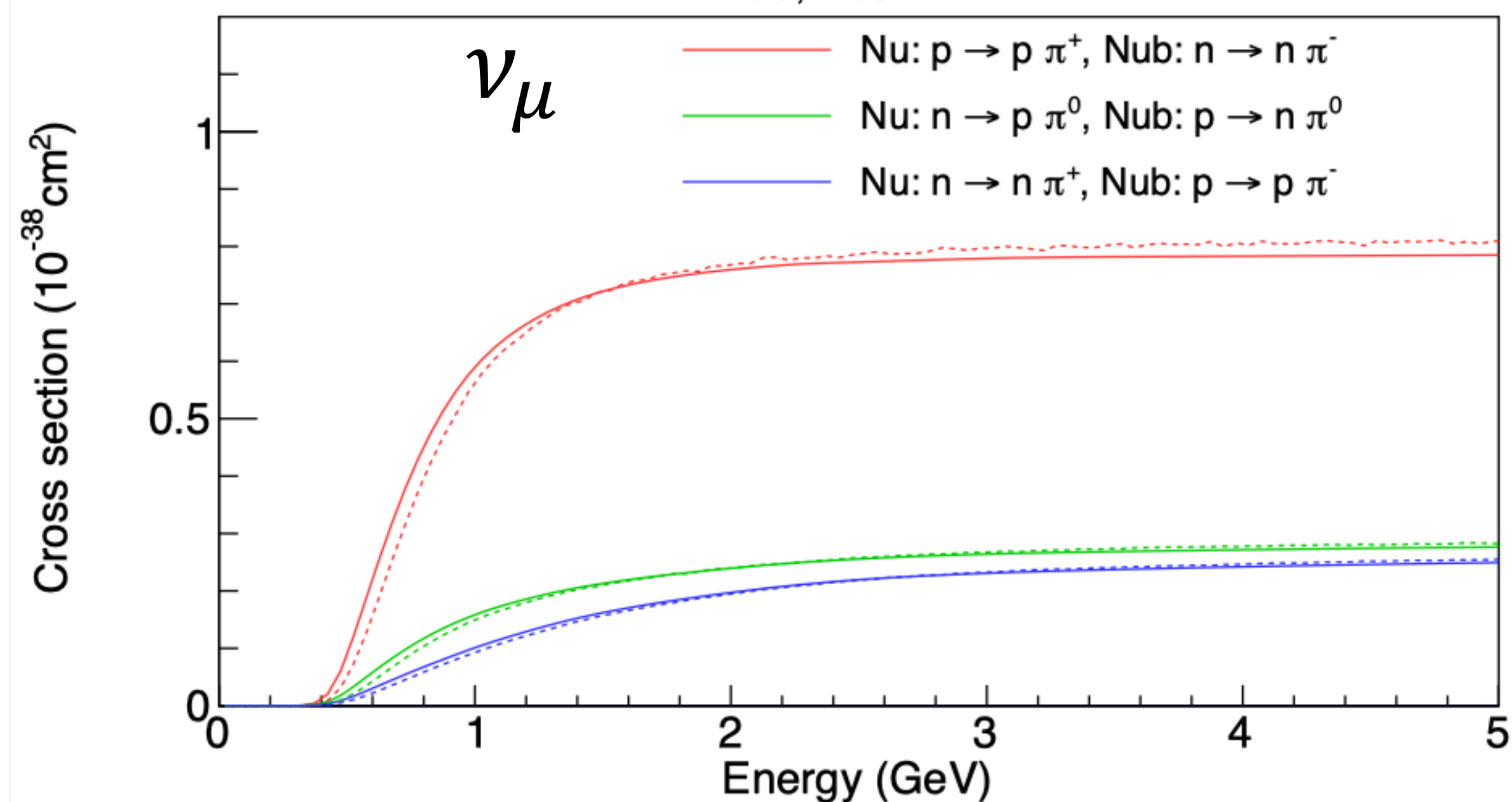


- SF has smaller Xsec than LFG with Eb.

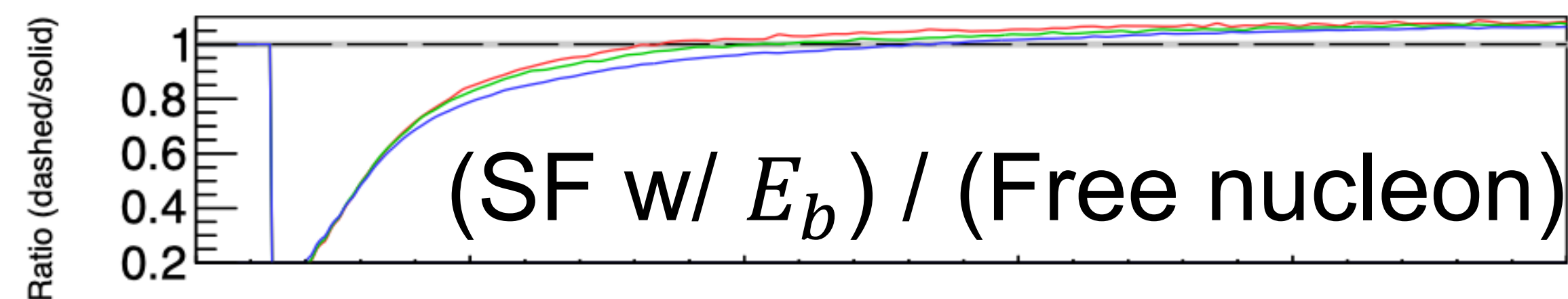
**LFG with Eb (dashed)
vs free nucleon (solid)**



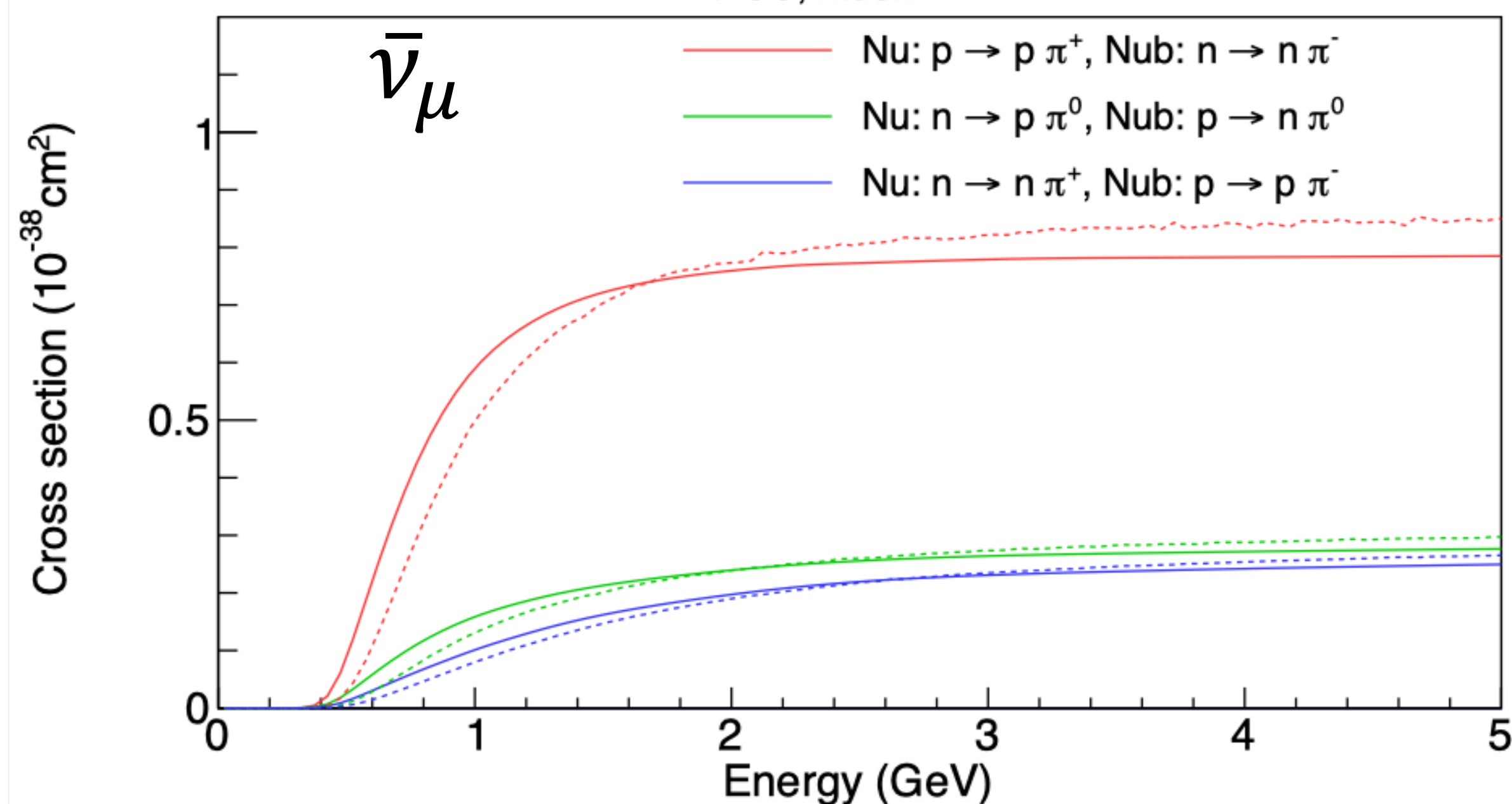
ν CC, muon



**SF with Eb (dashed)
vs free nucleon (solid)**



ν CC, muon



w/o E_b

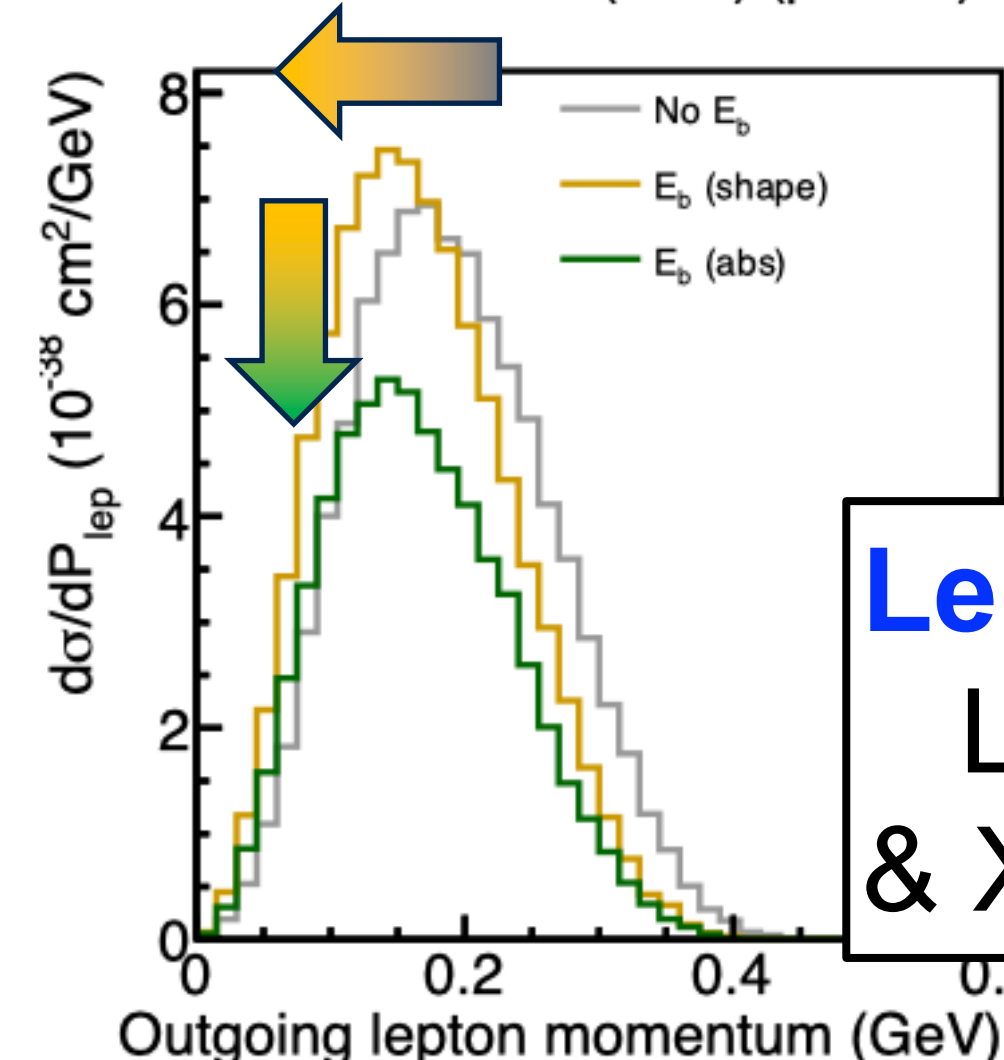
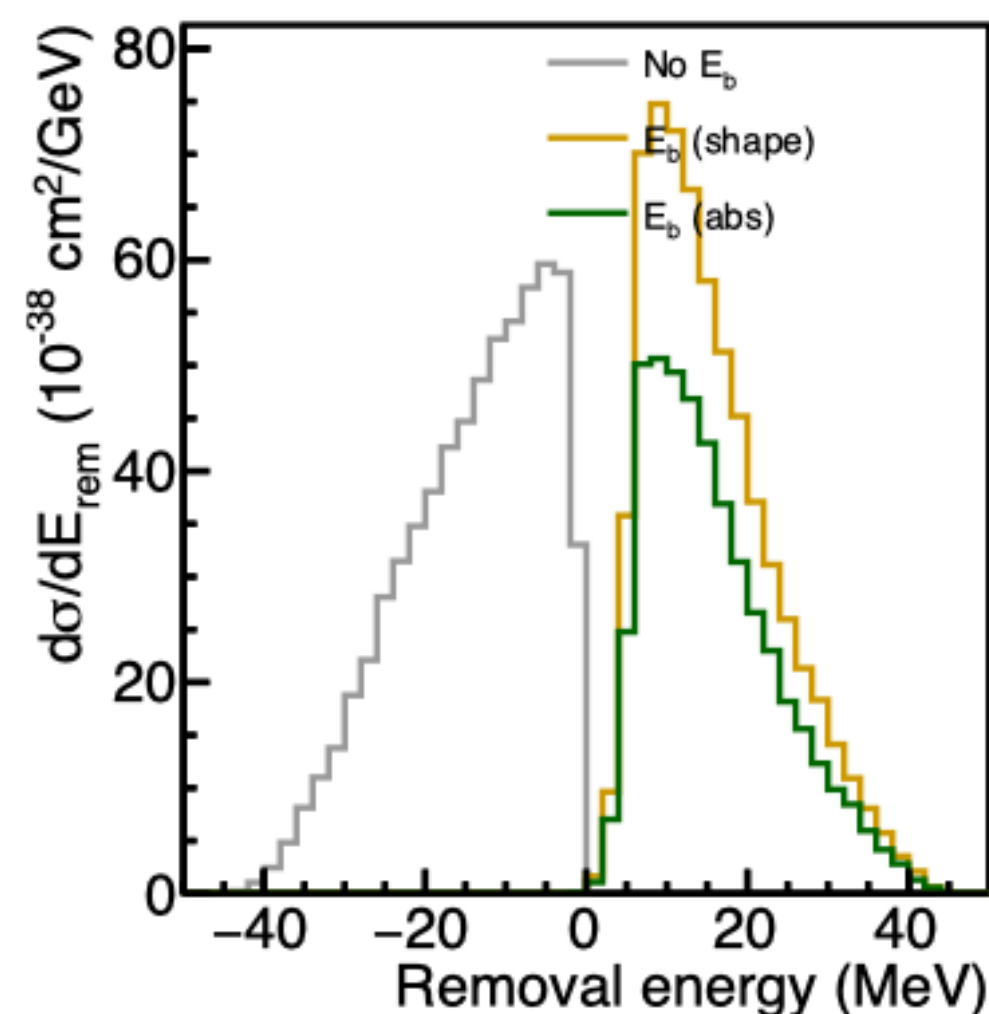
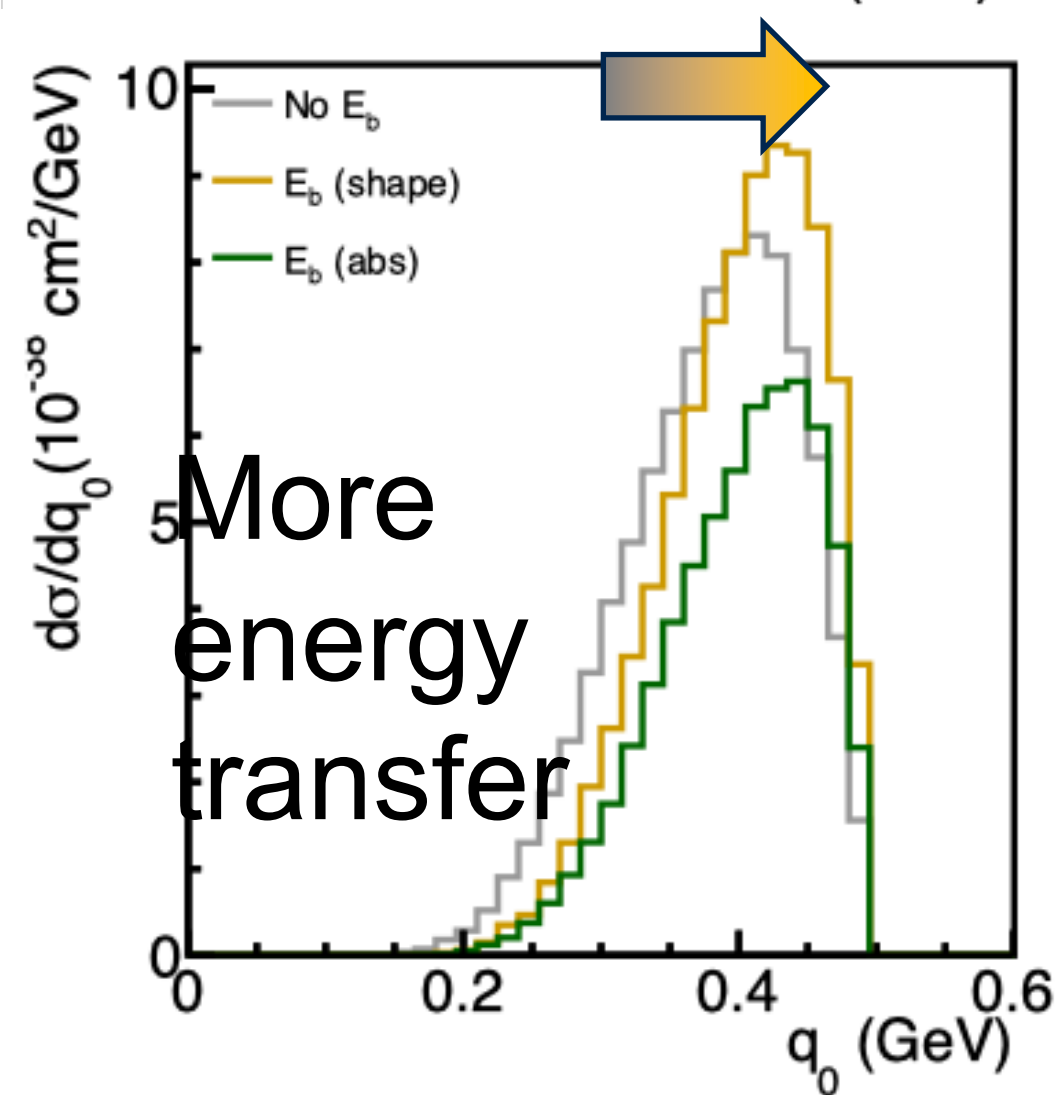
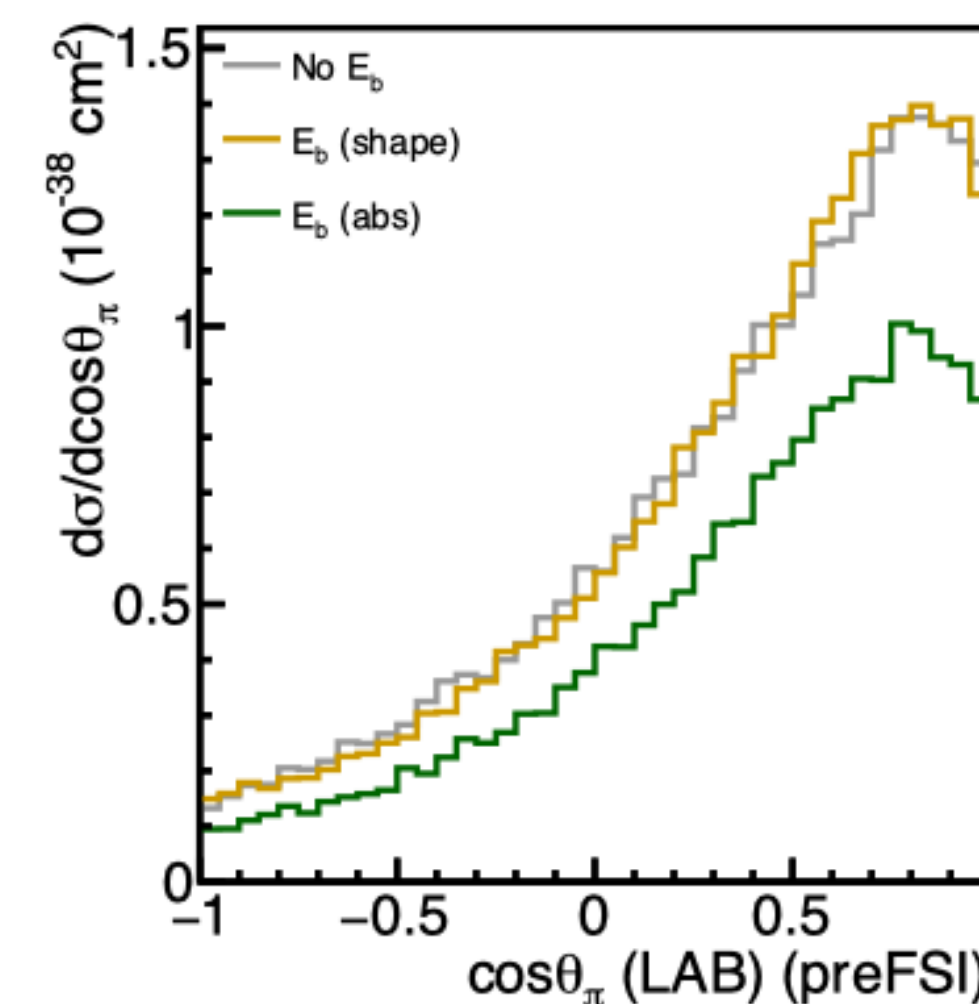
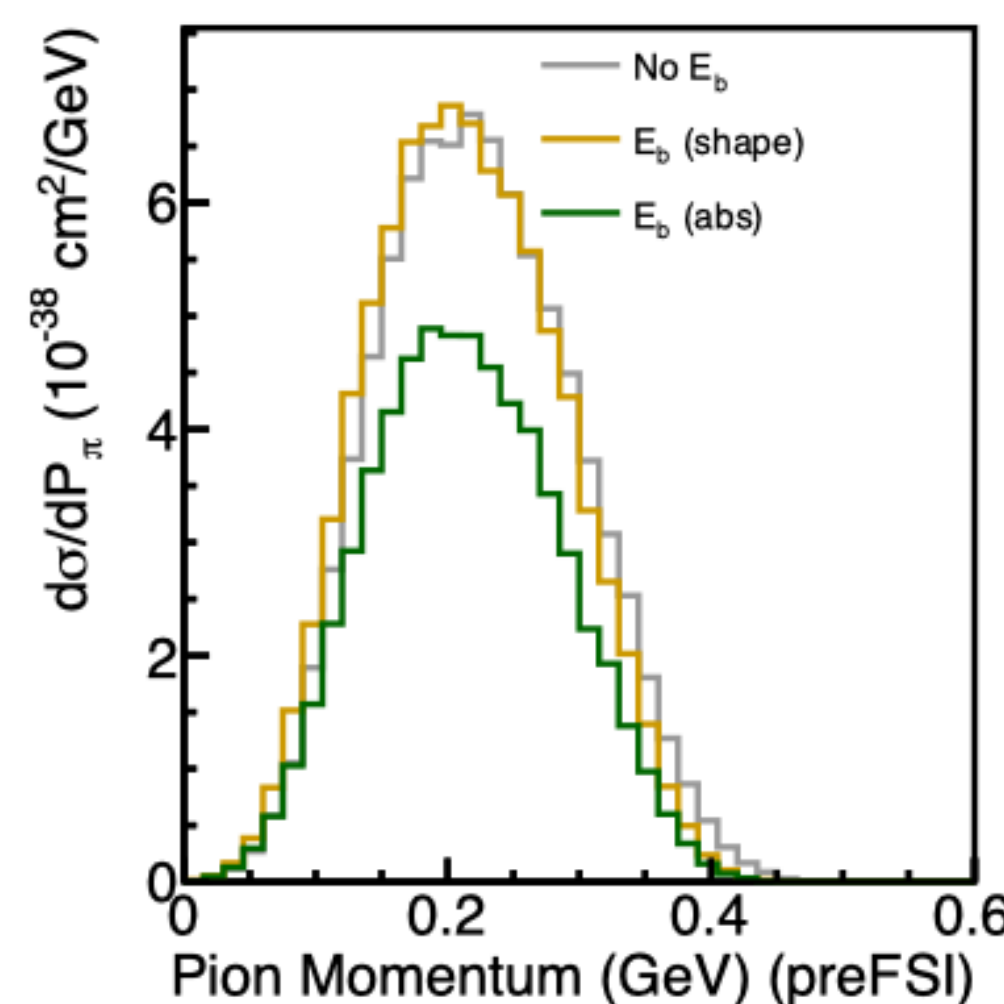
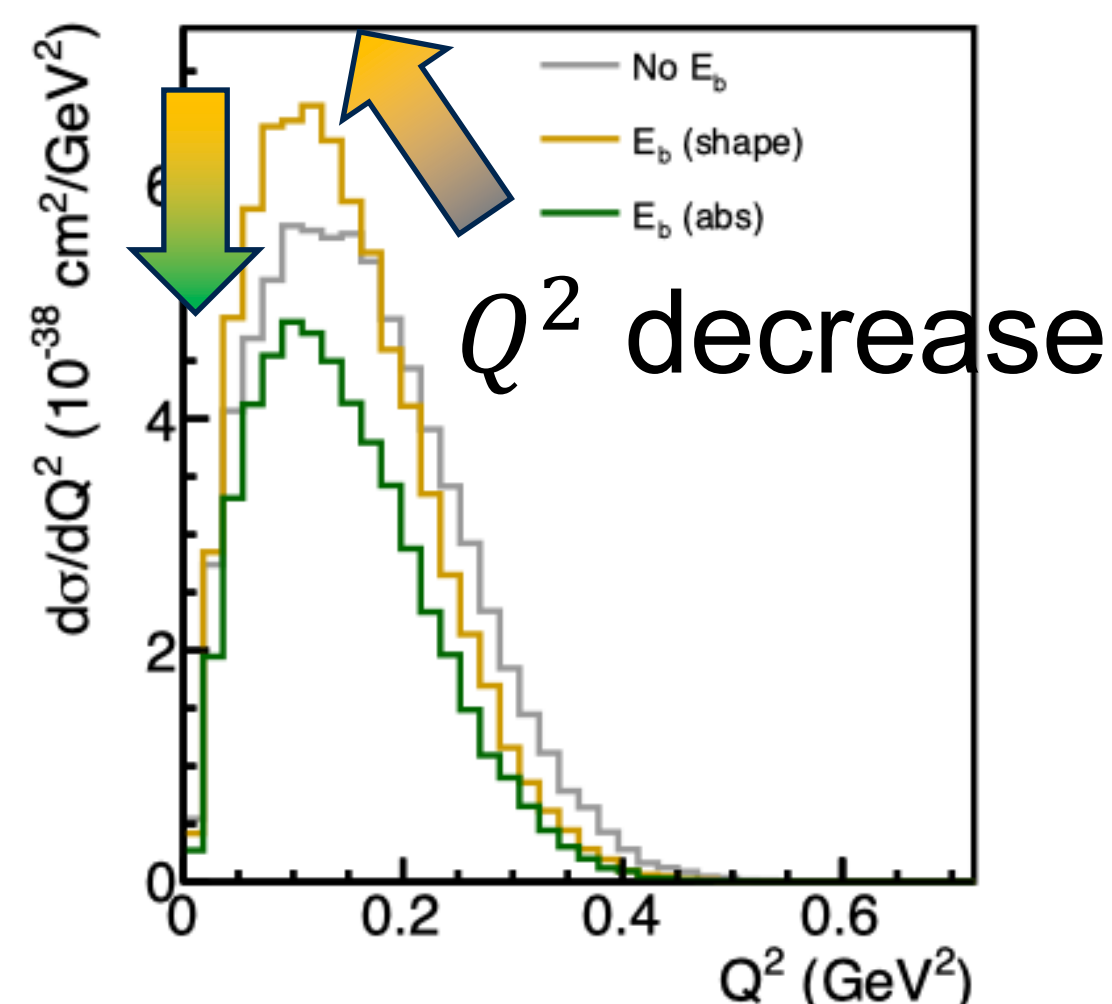
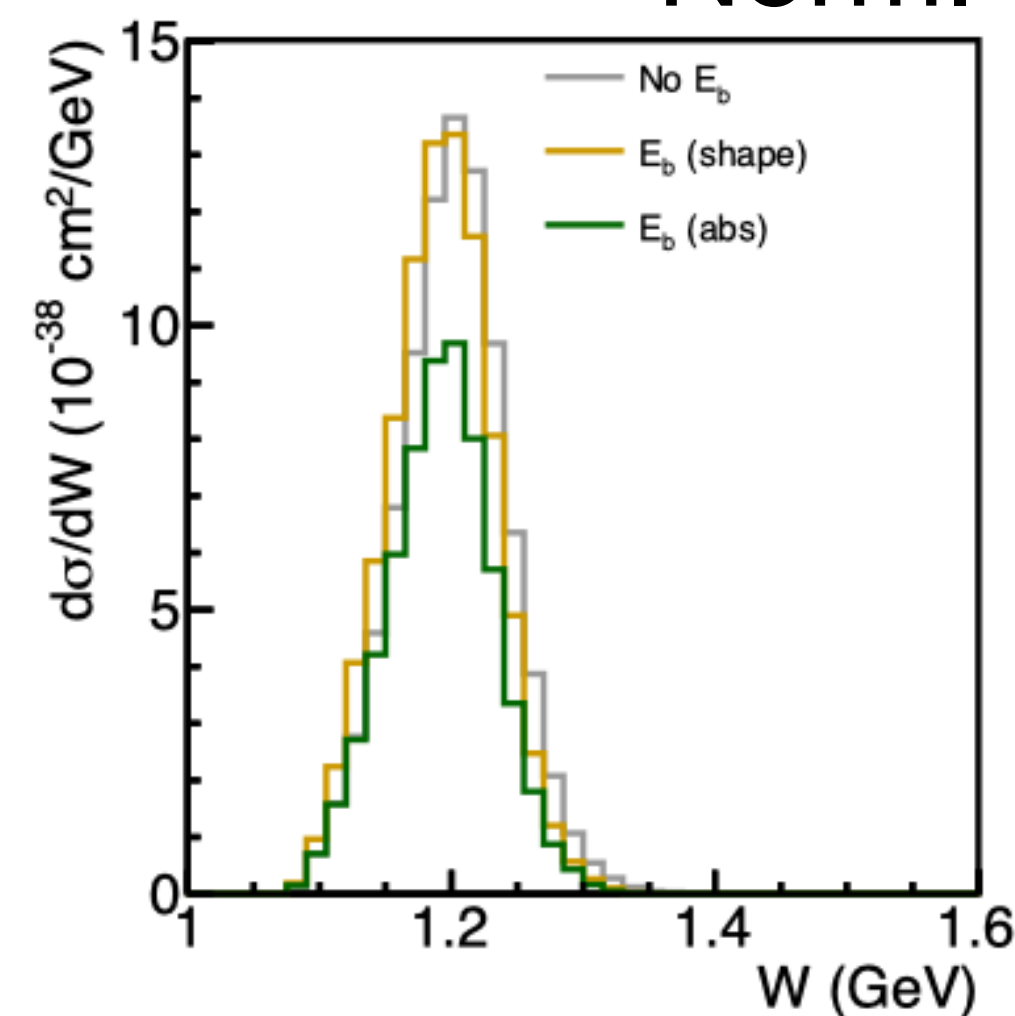
w/ E_b (normalized to w/o E_b)

w/ E_b

$\nu_\mu p \rightarrow \mu^- p \pi^+$ at 600 MeV.

LFG (RFG gives similar trend)

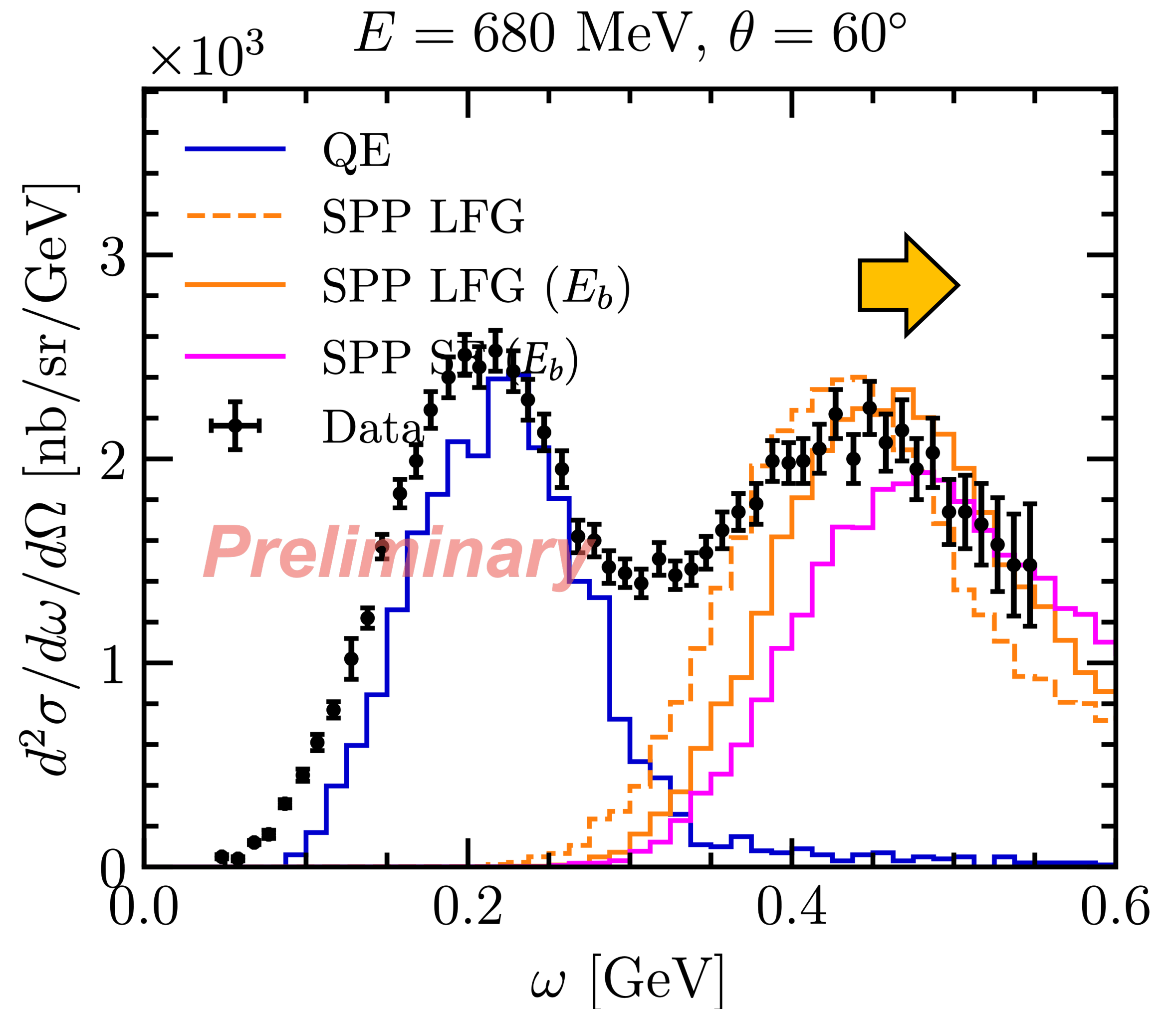
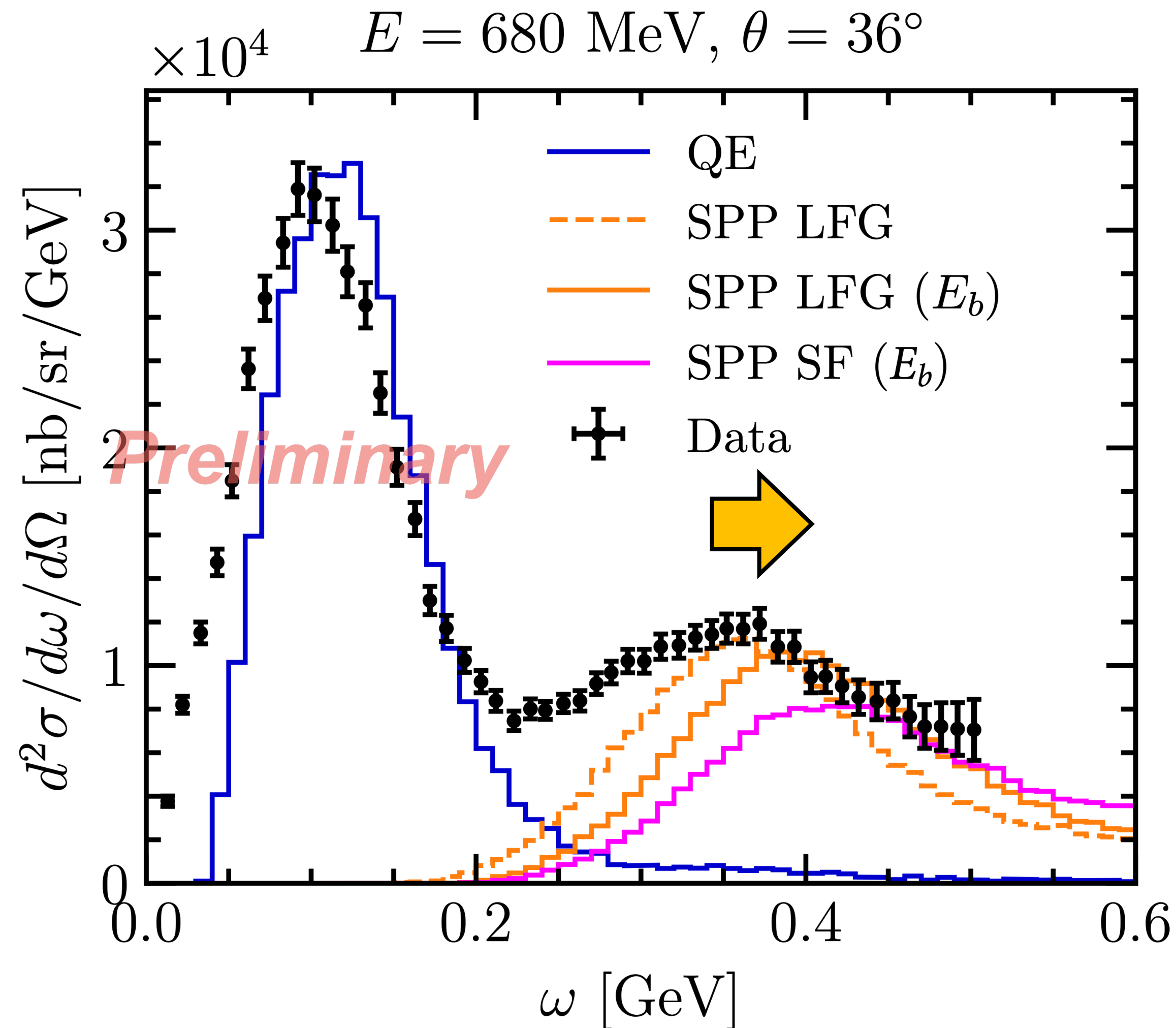
Norm. decrease



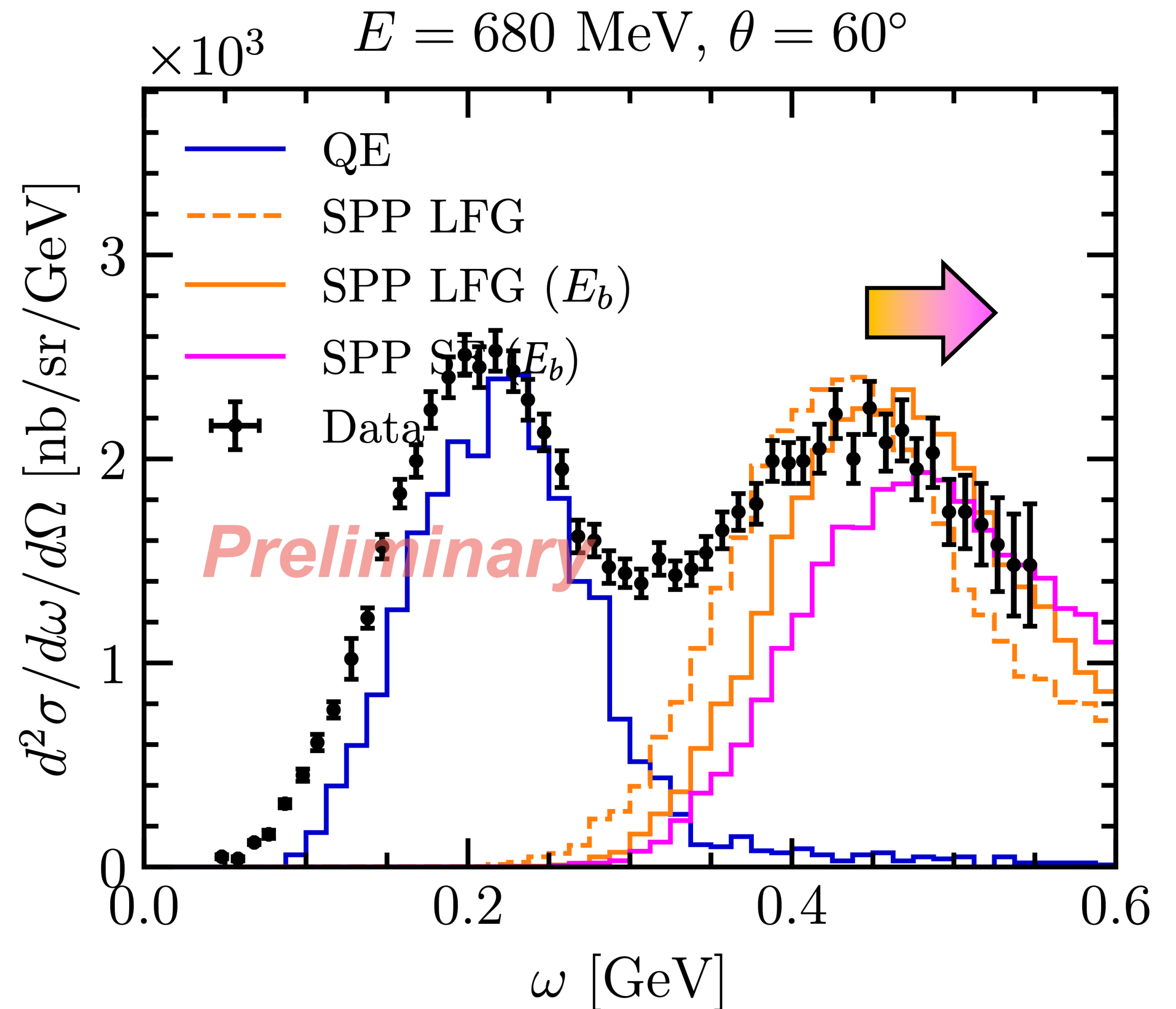
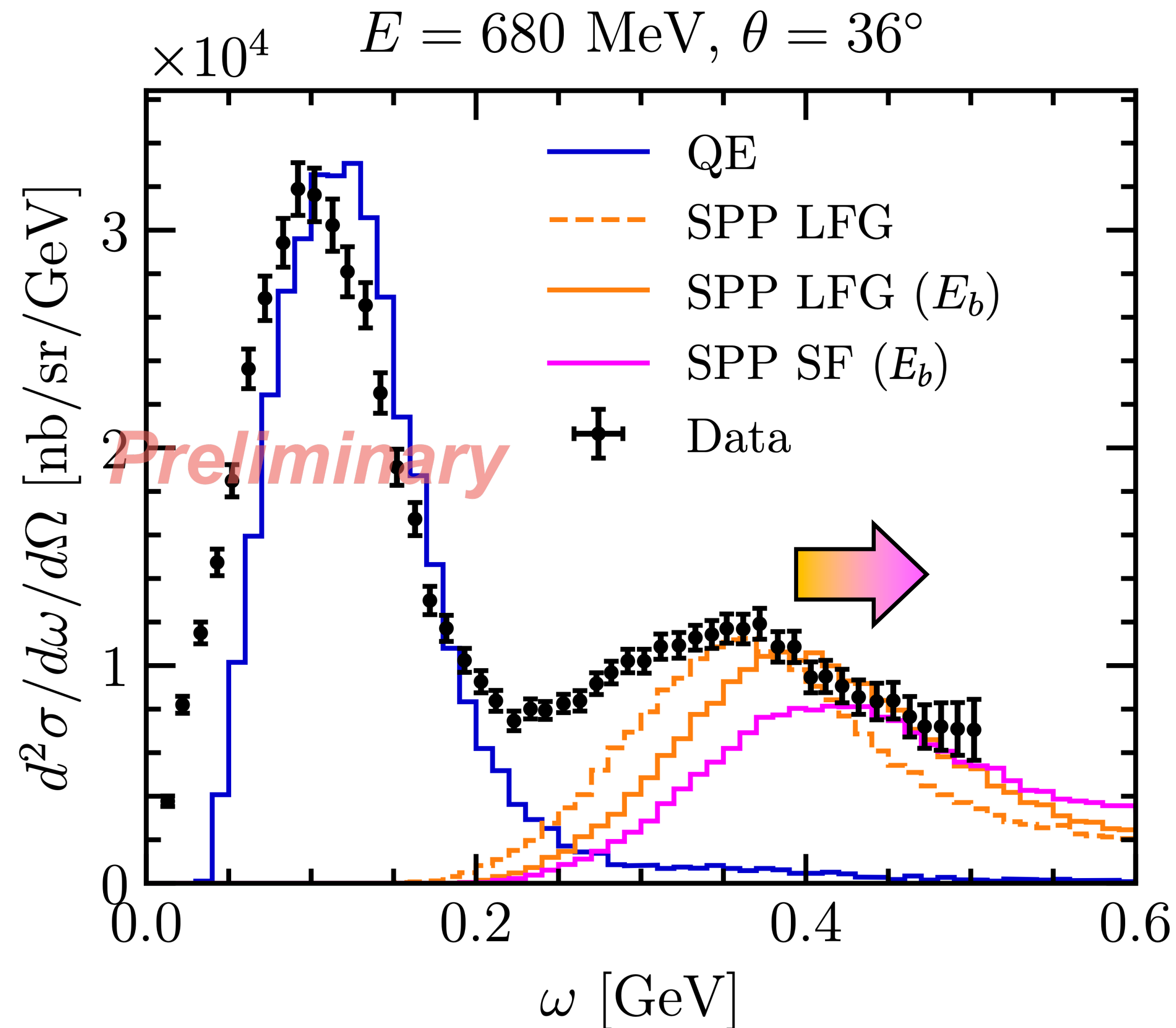
Pion has little change
in shape under this
nuclear model.

Lepton changes drastically:
Lower lepton momentum
& Xsec suppression by ~30%

- Binding energy E_b shifts the energy transfer ω peak toward the larger value.

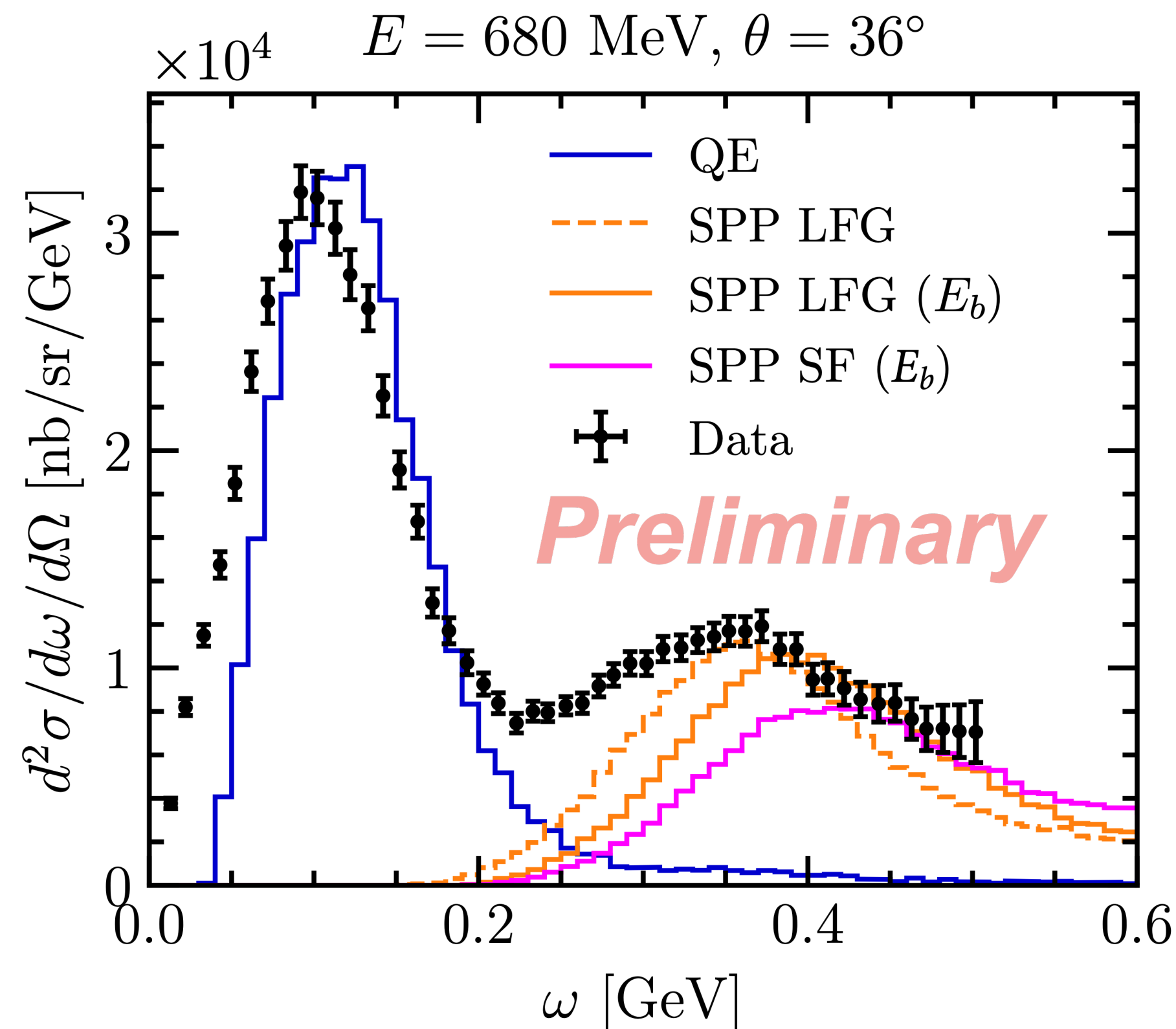


- ▶ Binding energy E_b shifts the energy transfer ω peak toward the larger value.
- ▶ SF peaks at more larger ω . Peak height become a little small.
- Roughly consistent behavior with Noemi et al., PRC 100, 045503 (2019).

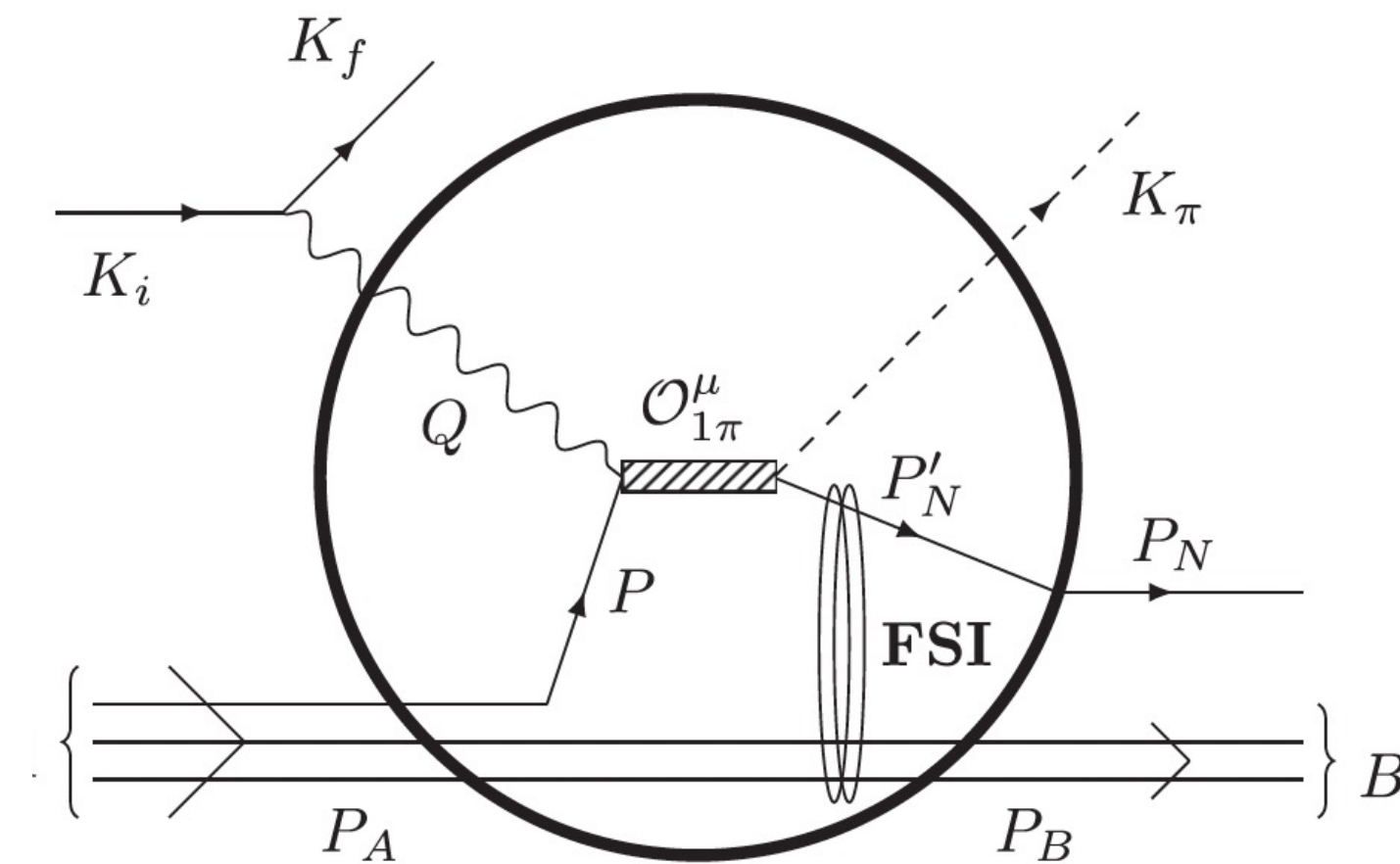
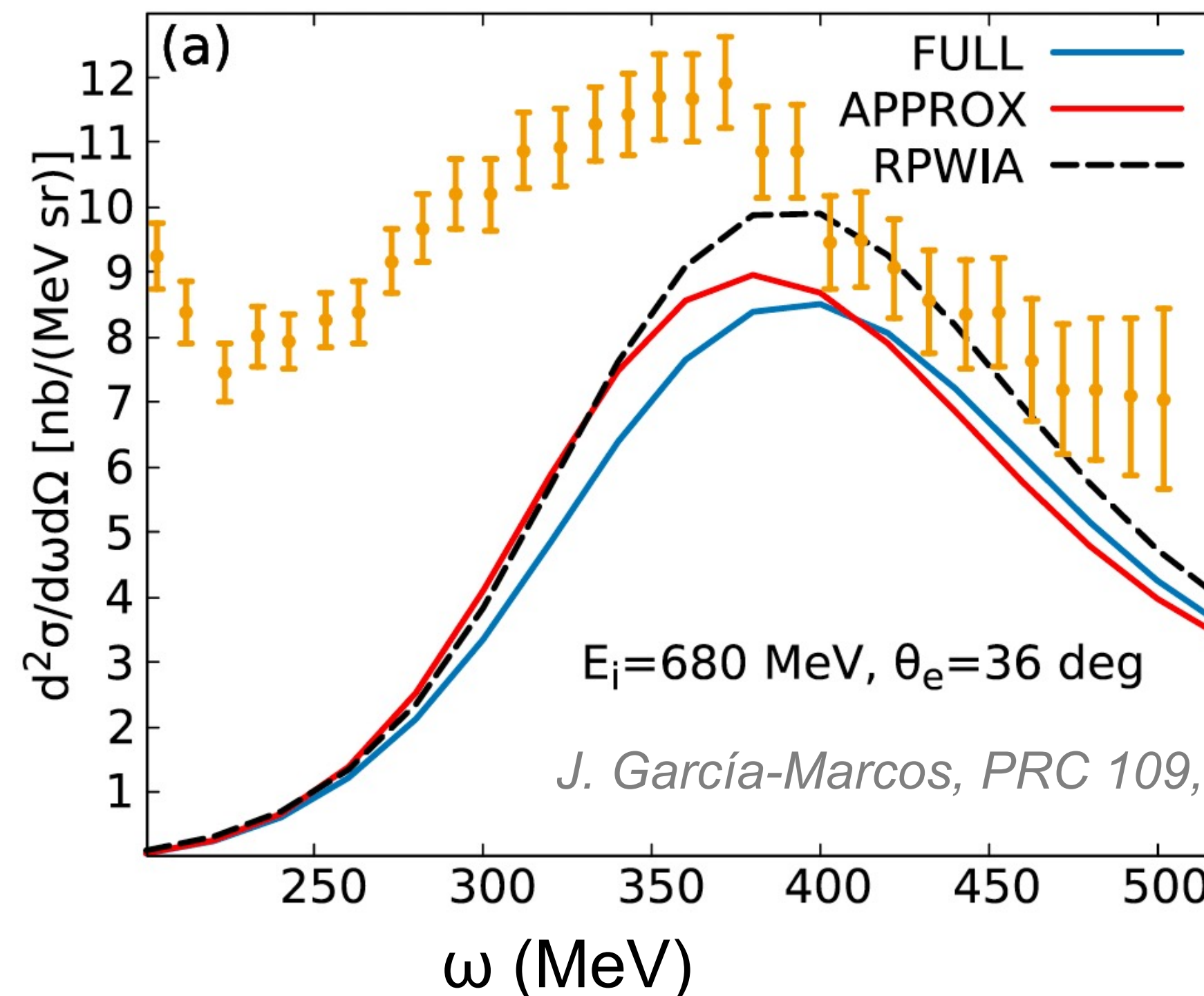


- Distortion and in-medium effects should give non-negligible impact, but are absent in NEUT.
- Is there any way to apply the distortion or in-medium effect to NEUT-DCC?
 - An ad-hoc method would be sufficient for the first step...?

NEUT simulation



c.f.) RDWIA based on ED-RMF



J. García-Marcos, PRC 109, 024608 (2024).

- DCC multipole amplitude is tabulated.
- **Therefore, it's impossible to evaluate model uncertainty by “reweight”.**
- **Start discussion how to achieve “reweight” in NEUT-DCC.**
 - The amplitude has several terms.
 - nonresonant, bare resonant, and meson-cloud.
 - By splitting these, part of reweighting is feasible.

Current

Total amplitude
tabulated

Differential Xsec

$$\frac{d^4\sigma}{dWdQ^2d\cos\theta_\pi^*d\phi_\pi^*}(E_\nu)$$

New (plan)

nonresonant

bare resonant

meson-cloud

tabulated

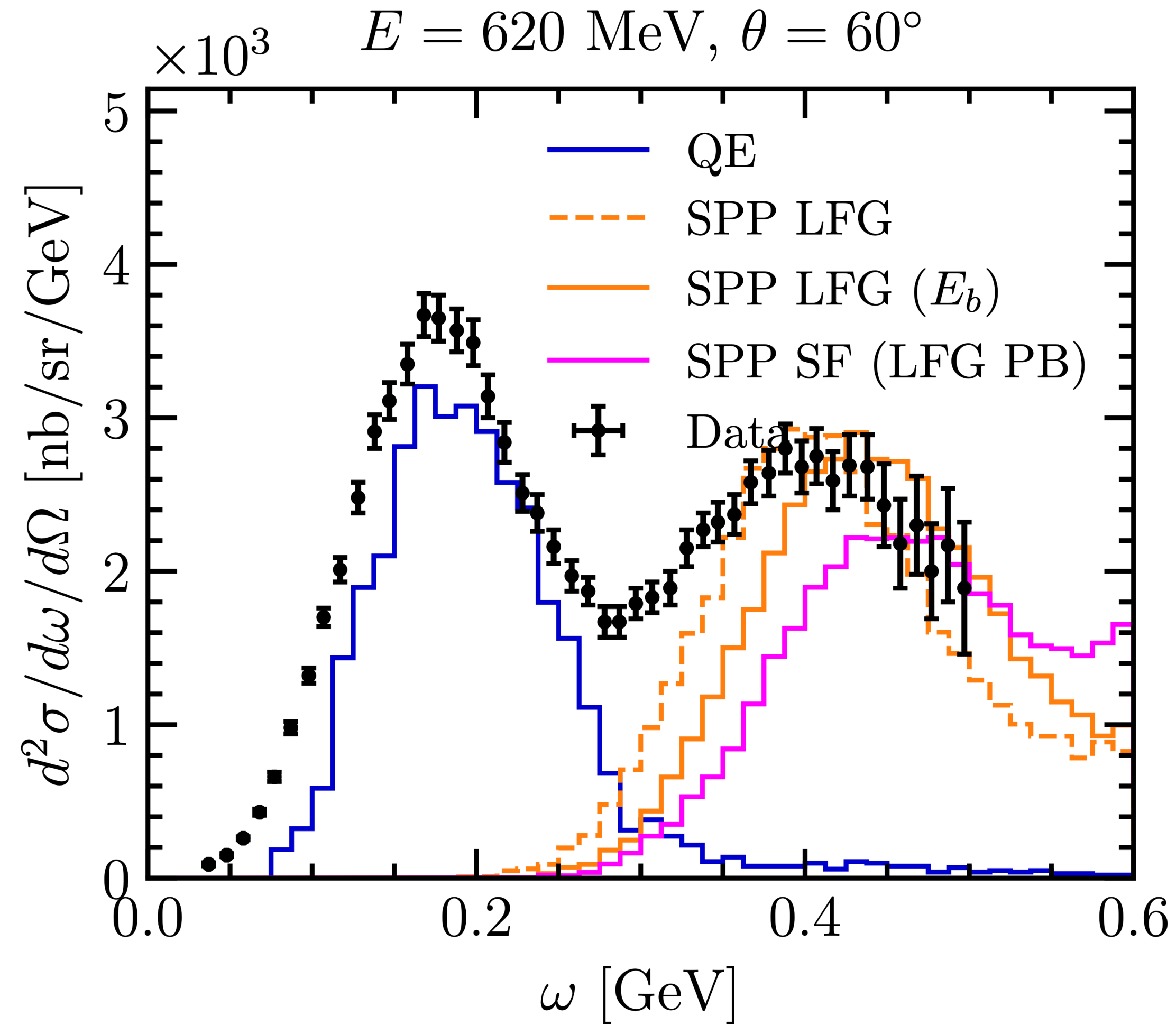
Total amplitude
on-the-fly

Change coupling
strength of resonances

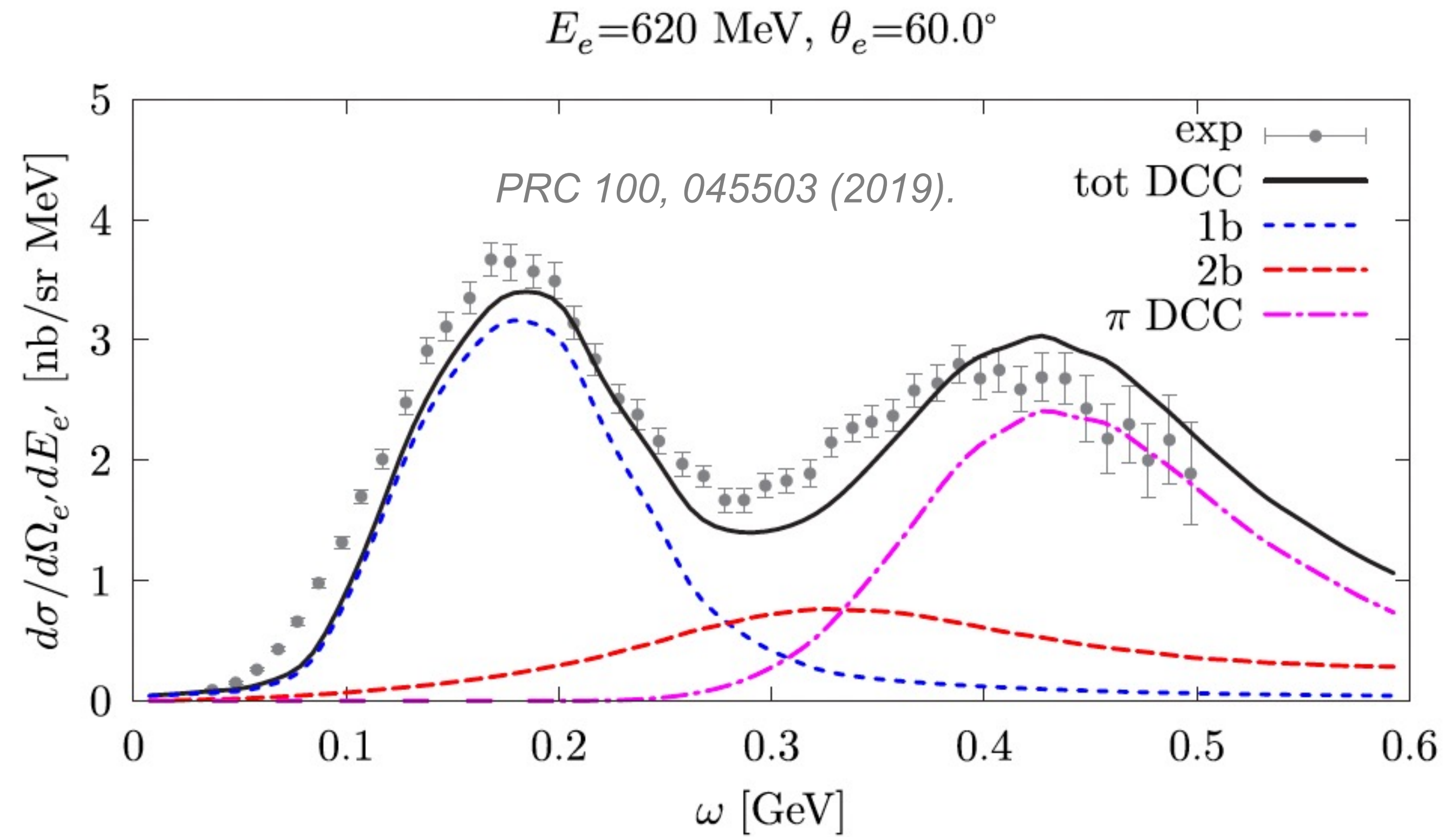
- NEUT has more SPP model choice rather than RS recently.
- **Full DCC model, including Adler angle sampling, was implemented.**
- **NEUT SPP models were based on naive nuclear models.**
- **Newly introduced binding energy, the same approach as NuWro.**
 - SF, LFG, and RFG are implemented.
 - Significant impact on lepton kinematics is observed.
 - **Impacts on SK/T2K/HK observables are to be investigated.**
- **There are big challenging topics!**
 - NEUT-DCC reweighting for uncertainty estimation.
 - Nuclear effects (distortion and in-medium effects of delta) for high-precision prediction.

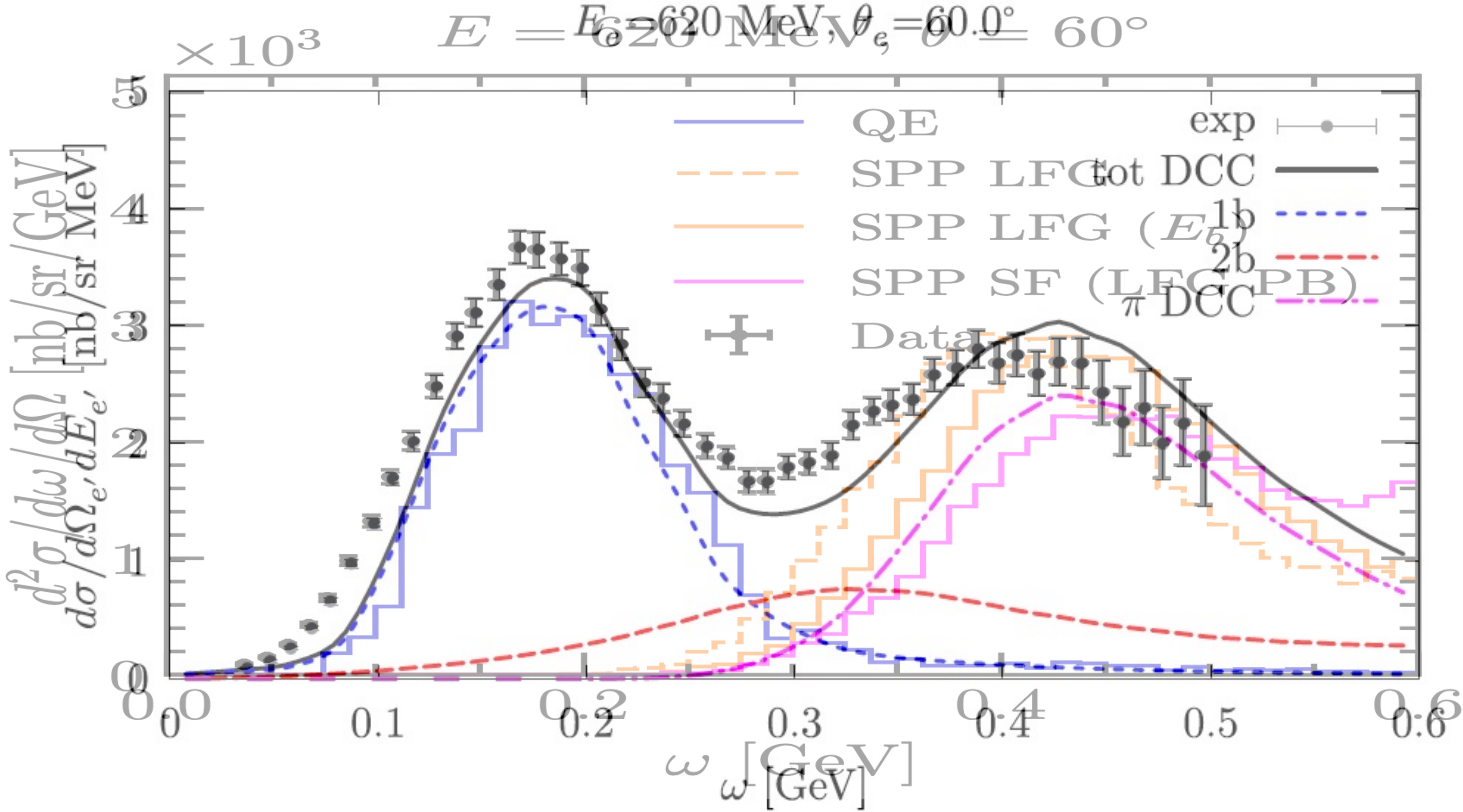
backup

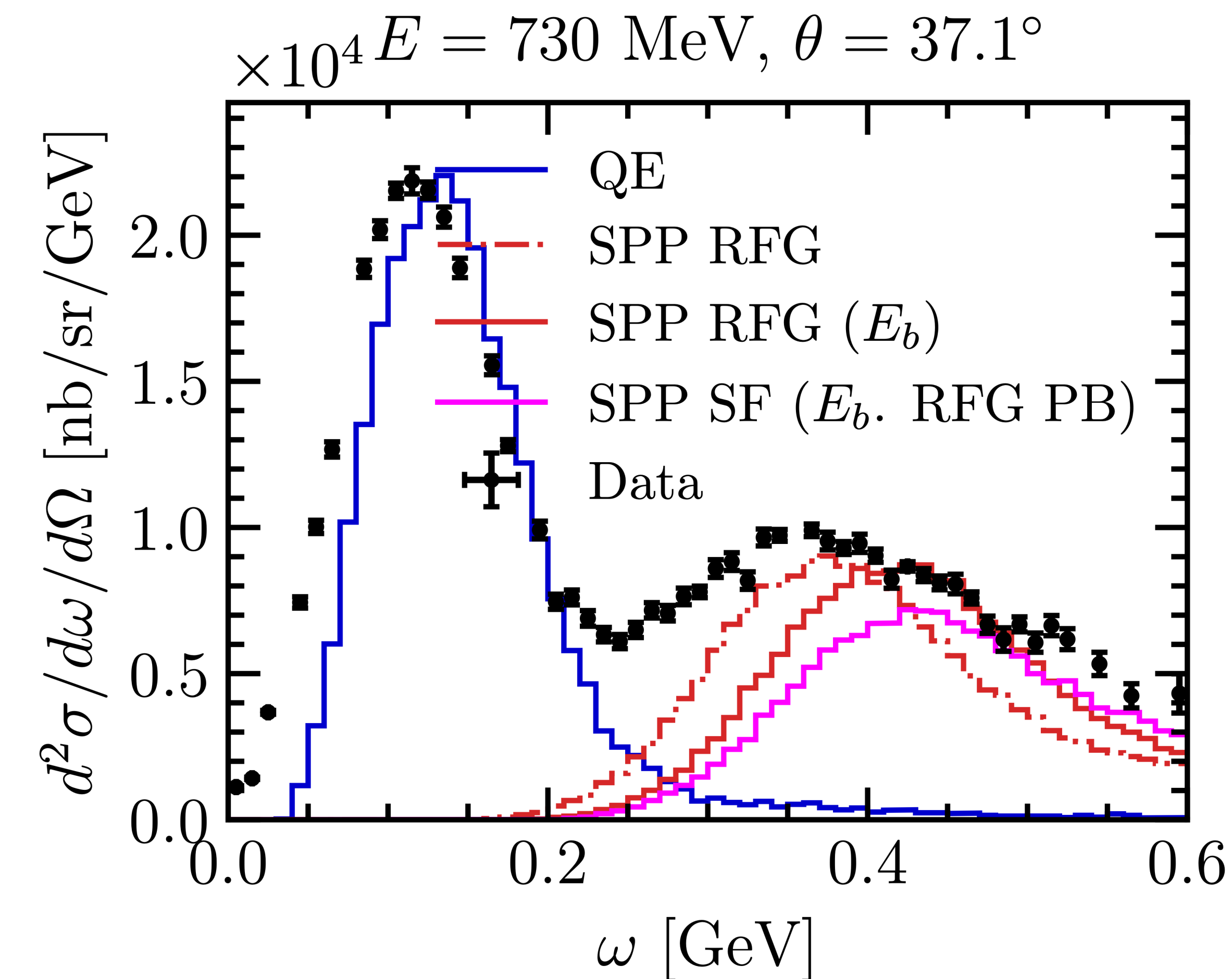
NEUT-DCC



N. Rocco et al. DCC with SF







N. Rocco et al. DCC.

$E_e = 730 \text{ MeV}, \theta_e = 37.0^\circ$

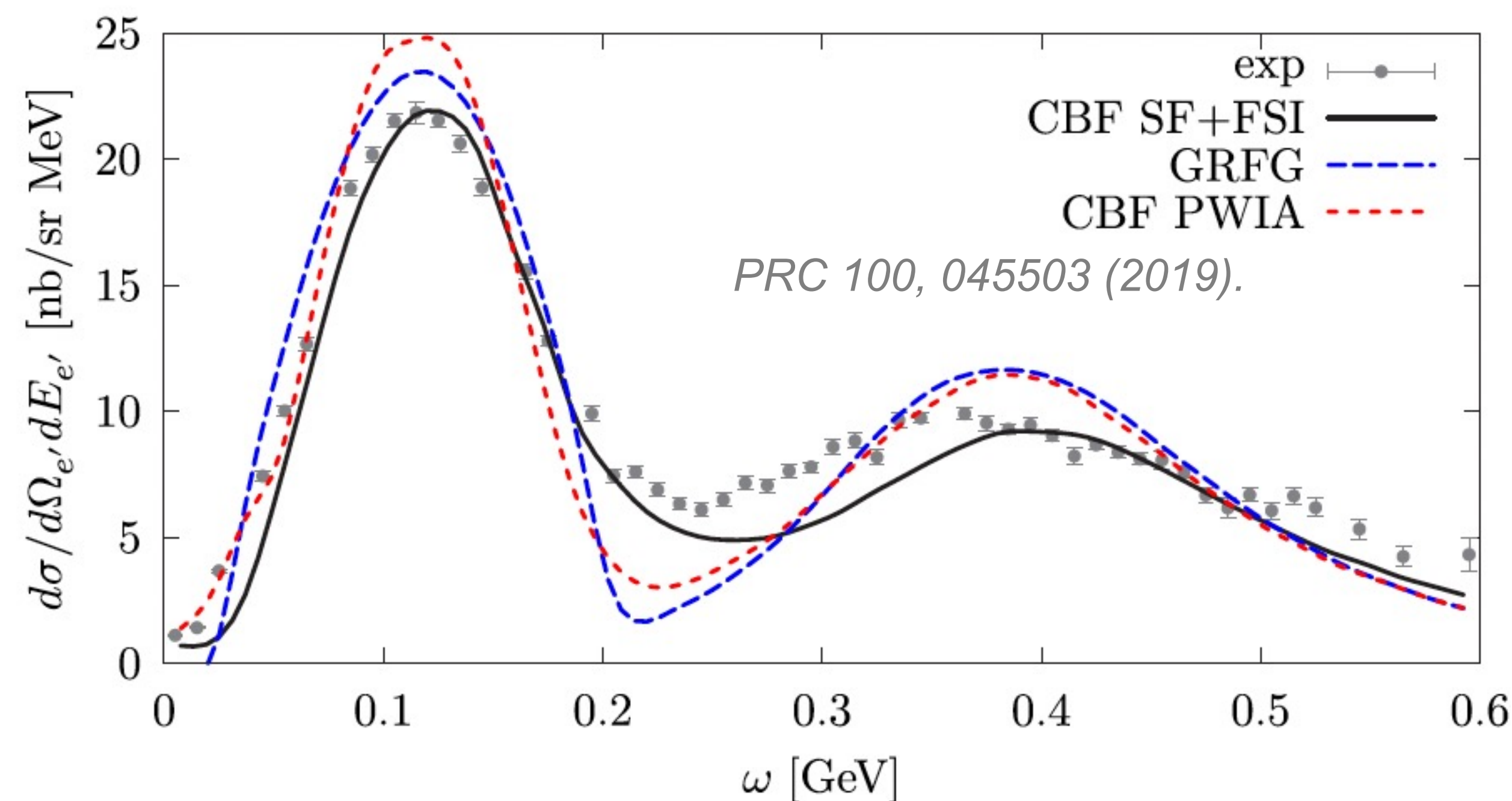


FIG. 10. Electron- ^{12}C double-differential cross section. The dashed (blue) line has been obtained within the GRFG model. The short-dashed (red) and solid (black) curves have been obtained using the SF of Ref. [39] within the PWIA and IA with FSI corrections, respectively.

RFG w/o E_b

SF PWIA (probably QE only)